
AURA: Visually Interpretable Affective Understanding via Robust Archetypes

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Abstract

Interpretable methods such as vision–language models (VLMs) have advanced affect analysis by aligning images with textual descriptions. However, relying on text as an intermediate proxy faces critical limitations: linguistic templates are inherently discrete, making them fundamentally incompatible with continuous valence–arousal regression, while also acting as a bottleneck for fine-grained visual nuances. Cognitive psychology suggests that human affective perception is not mediated by linguistic translation, but is grounded in perceptual resemblance to internalized visual archetypes. Motivated by this, we propose AURA, an archetype-based framework that replaces brittle linguistic proxies with a self-organizing archetype manifold. By adaptively allocating representational density according to affective complexity, AURA enables accurate continuous regression and reshapes affective taxonomies by decomposing labels into interpretable, geometrically coherent visual primitives. This paradigm offers a transparent, visually grounded decision trail and achieves state-of-the-art results across discrete and continuous tasks.

1. Introduction

Recent affective computing studies have increasingly emphasized interpretable, robust, and multimodal representations for human-centered affect understanding (Hu et al., 2025c; Kollias et al., 2024). In parallel, large-scale vision–language models, such as CLIP (Radford et al., 2021), have reshaped representation learning by aligning visual inputs with broad semantic concepts. This capability has recently been leveraged to advance text-driven interpretable affective understanding, particularly for Facial Expression Recogni-

tion (FER) and Action Unit (AU) detection (Zhou et al., 2022b; Li et al., 2024a; Hu et al., 2026; Liu et al., 2025).

However, despite these advances, the prevailing text-dependent paradigm faces four critical limitations that impede its applicability to fine-grained affect modeling: **(i) Incompatibility with Continuous Regression.** Dimensional affective representations, such as the Valence–Arousal (VA) space, are essential for fine-grained analysis. However, the classification-oriented design of VLMs, which relies on discrete text–image matching, is fundamentally ill-suited for modeling continuous regression manifolds. **(ii) Reliance on Brittle Linguistic Prompts.** Current approaches attempt to encapsulate inherently fuzzy affective states within linguistic templates (Zhou et al., 2022b). For FER, some methods even design elaborate multi-sentence descriptions for particular facial expressions (Zhao & Patras, 2023; Li et al., 2024a), reducing rich visual patterns to finite linguistic descriptions that still struggle to cover the large intra-expression diversity. Similarly, in AU detection, complex Facial Action Coding System (FACS) codes (Ekman & Friesen, 1978) are often translated into plain natural-language descriptions (e.g., “AU5: Upper Lid Raiser” → “wide-open eyes”), reducing precise anatomical definitions to coarse linguistic approximations (Chang et al., 2024; Liu et al., 2025). **(iii) Fragmentation of Affective Tasks.** Existing affective analysis methods are often organized around task-specific formulations, with FER, AU detection, and VA regression handled by separate architectures. This task-bound design limits the emergence of a general learning paradigm that can induce structured and interpretable visual representations for different affective tasks. **(iv) Computational Inefficiency.** Adapting VLMs to affective tasks often requires fine-tuning large visual or textual encoders, leading to high computational overhead. Such adaptation can further specialize a general-purpose VLM to a single affective task, weakening its transferable representation capacity.

More fundamentally, text-dependent approaches diverge from the cognitive mechanisms of human emotional perception. Cognitive psychology suggests that affective understanding is not necessarily mediated by an intermediate linguistic translation (Facial → Linguistic Description → Perception), but can be approached through *perceptual resemblance to internalized visual archetypes* (Facial → Visual Archetype → Perception) (Rosch, 1975; Fehr &

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Russell, 1984; Nosofsky, 1986; Shepard, 1987; Barsalou, 1999). Carl Jung characterized archetypes as primordial, recurring patterns that shape perception and meaning (Jung, 1968). **We reinterpret this view geometrically:** a visual archetype corresponds to a *local visual semantic extreme* that anchors a localized semantic region on the visual manifold, such that a collection of archetypes yields a structured decomposition of affect into *interpretable subregions* with smooth transitions between them. In this sense, affective perception can be viewed as locating a facial instance within this archetype manifold and reading out affect from the archetypal neighborhood it activates, with the selected archetype serving as the immediate semantic basis of the judgment. Motivated by this cognitive view, we introduce **AURA** (Affective Understanding via Robust Archetypes), a *visually interpretable* framework that operationalizes this archetype-based perception to support regression, classification, and detection within a general learning paradigm. Eliminating reliance on textual prompts or encoder fine-tuning, AURA projects frozen pre-trained visual embeddings into an archetype space to reconstruct the intrinsic affective geometry. Crucially, the framework adaptively discovers and refines the necessary semantic extremes, populating the manifold with a dense set of local archetypes that *self-organize* according to data complexity. Consequently, prediction is reduced to an interpretable measurement of resemblance between the projected embedding and the archetypes.

Notably, AURA departs fundamentally from conventional prototype- or clustering-based representation learning methods (Snell et al., 2017; Hu et al., 2025b; Xie et al., 2016; Caron et al., 2018). Such methods organize embeddings around predefined representative units, such as class prototypes, class means, or cluster centroids. This representative-centric formulation imposes a coarse discretization on the affective manifold, obscuring fine-grained visual modes, ambiguous transitions, and continuous intensity variations. Specifically, they face three persistent hurdles: **(i) Static Representational Cardinality.** Existing methods usually prescribe the number of representative units in advance, leaving their representational capacity insensitive to the uneven complexity of affective data and inadequate for visually diverse or high-variance regions. **(ii) Representative Homogenization.** By summarizing samples through central representatives, these methods can compress distinctive local modes into coarse semantic abstractions, thereby diluting the fine-grained cues needed to distinguish subtle, compound, or ambiguous affective patterns (e.g., melancholy vs. neutral). **(iii) Topological Rigidity.** These representative units are typically optimized for discrete recognition or grouping objectives, encouraging separated partitions rather than a continuous affective topology and limiting smooth semantic transitions required by continuous regression tasks.

In contrast, AURA addresses these limitations through a

flexible, geometry-preserving learning framework: (i) it **adaptively discovers** a stable set of active archetypes, allocating denser representations to visually complex or high-variance affective states; (ii) it operates via **sharp semantic anchors** (local extremes) rather than diffuse averages, thereby preserving fine-grained interpretability; and (iii) it provides a general **archetype-based modeling principle** that can be instantiated across diverse affective tasks. Accordingly, AURA adapts the archetype topology to the intrinsic structure of each task, supporting discrete classification, fine-grained detection, and continuous regression. A detailed comparison is provided in Appendix D.

Finally, AURA advances semantic-level interpretability by exposing the structural logic behind each prediction. It provides a decision trail by identifying **which** archetype anchors the input, **why** it is selected, and **what** canonical visual pattern it represents. Moreover, such archetype-level reasoning enables **dataset-level diagnosis**, revealing potential annotation inconsistencies, ambiguous samples, and taxonomy refinements through the lens of archetypes.

2. AURA Framework

To operationalize archetype-based perception, AURA reconstructs the intrinsic affective geometry through three components, as illustrated in Fig. 1. First, a **Visual Archetype-Space Projector** maps frozen visual embeddings into a compact learnable manifold, establishing a geometric basis for archetype discovery. Within this space, the **Self-organized Archetype Discovery Module** introduces *Discriminative Assignment* and *Adaptive Regularization* to populate the manifold with local semantic anchors, thereby self-organizing representational density according to affective complexity. Finally, the **Archetype Contextualization Module** enforces structural coherence via *Distance-Aware Contextualization*, refining archetype neighborhoods to support semantic transitions.

2.1. Visual Archetype-Space (VAS) Projector

To bridge generic visual representations and task-specific affective semantics, the VAS Projector maps frozen high-dimensional visual embeddings into a compact learnable subspace, which serves as the *visual archetype manifold*. To dissociate emotional expressions from subject-specific and environmental confounds, this projection *distills the affective essence* by steering embeddings toward an archetype-centric geometry, suppressing affect-irrelevant variations, and emphasizing discriminative facial cues. To accommodate the heterogeneity of affective perception, VAS further adapts hierarchical features according to task-specific granularity, extracting global cues for holistic tasks while preserving patch-level details for localized AU detection. The VAS space provides the geometric basis for archetype discovery.

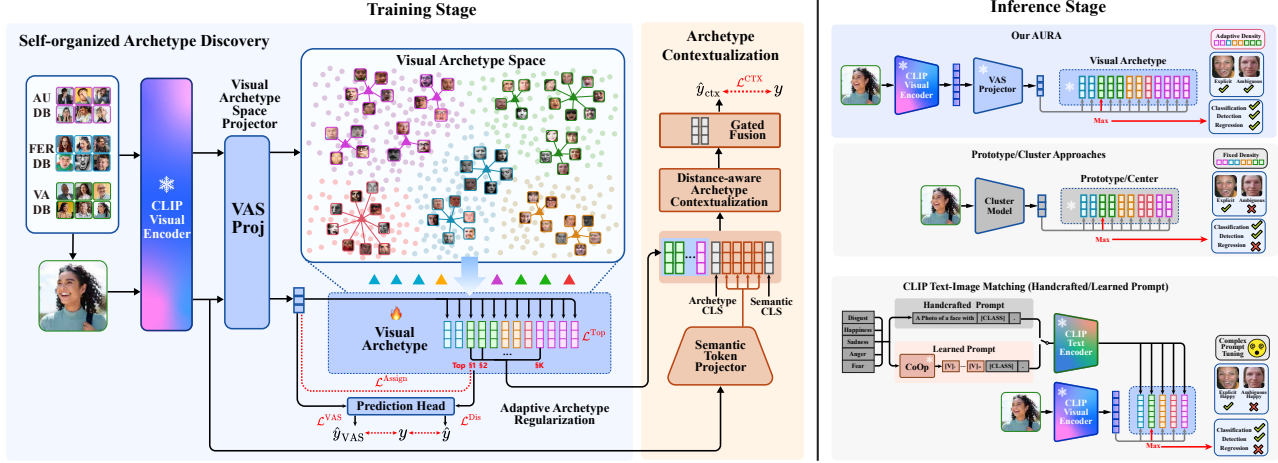


Figure 1. Overview of AURA. **Left: Training stage.** AURA supports FER, AU detection, and VA regression through three components: a *Visual Archetype-Space Projector* that maps frozen CLIP features into a task-specific archetype space, *Self-organized Archetype Discovery* that learns adaptive-density visual archetypes, and *Archetype Contextualization* that refines local semantic structure through distance-aware neighbor reasoning. **Right: Inference stage.** AURA performs lightweight perceptual read-out by projecting a visual feature into the archetype space and matching it to fixed archetypes via cosine similarity. **Comparison.** Prototype/cluster methods typically rely on a fixed number of representative centers, while CLIP text–image matching depends on handcrafted or learned prompts. In contrast, AURA learns prompt-free visual archetypes with adaptive density, providing fine-grained and interpretable affective understanding.

(i) Global-level Embedding. Holistic affective tasks (e.g., FER and VA) rely on global facial cues. Given an image x , we extract the global embedding from the frozen CLIP visual encoder as $\mathbf{z}_g = \mathcal{E}_{\text{CLS}}(x) \in \mathbb{R}^D$. The embedding is projected into the archetype manifold via a lightweight task-specific projector to obtain $\mathbf{f}_\tau = \mathcal{F}_\tau(\mathbf{z}_g) \in \mathbb{R}^d$, where τ denotes the task and $d \ll D$. To ensure the projection space captures discriminative affective signals while suppressing irrelevant variations, we impose task-specific supervision.

For FER, we use classifier $\phi_{\text{Exp}}(\cdot)$ with class-weighted cross-entropy, where $w_y = \frac{N}{Cn_y}$ for expression label y , total samples N , classes C , and class size n_y :

$$\mathcal{L}_{\text{Exp}}^{\text{VAS}} = -w_y \log \phi_{\text{Exp}}(\mathbf{f}_{\text{Exp}})_y. \quad (1)$$

For VA regression, distinct heads $\phi_V(\cdot)$ and $\phi_A(\cdot)$ operate on VA features. We denote batch-level predictions as $\hat{\mathbf{y}}_V = \phi_V(\mathbf{f}_V)$ and $\hat{\mathbf{y}}_A = \phi_A(\mathbf{f}_A)$, with targets $\mathbf{y}_V$ and $\mathbf{y}_A$. The objective combines Concordance Correlation Coefficient (CCC) with MSE to align prediction trends and values:

$$\mathcal{L}_{\text{VA}}^{\text{VAS}} = 1 - \frac{1}{2} \left[\text{CCC}(\hat{\mathbf{y}}_V, \mathbf{y}_V) + \text{CCC}(\hat{\mathbf{y}}_A, \mathbf{y}_A) \right] + \frac{1}{2} \left[\text{MSE}(\hat{\mathbf{y}}_V, \mathbf{y}_V) + \text{MSE}(\hat{\mathbf{y}}_A, \mathbf{y}_A) \right]. \quad (2)$$

(ii) Patch-level Embedding. Conversely, AU detection targets localized muscle activations, necessitating fine-grained patch features. We extract patch embeddings $\mathbf{z}_p \in \mathbb{R}^{L \times D}$ from the final encoder layer. A lightweight Transformer $\mathcal{T}_{\text{proj}}(\cdot)$ aggregates these features with a learnable [CLS] token and positional encodings, and we take the output [CLS] representation as $\tilde{\mathbf{z}}_p \in \mathbb{R}^D$. To support multi-label detection, an independent projector $\mathcal{F}_{\text{AU}}^{(m)}(\cdot)$ maps $\tilde{\mathbf{z}}_p$ into

the AU-specific archetype space, yielding $\mathbf{f}_{\text{AU}}^{(m)} = \mathcal{F}_{\text{AU}}^{(m)}(\tilde{\mathbf{z}}_p)$. The feature is passed to an AU-specific detection head and optimized via binary cross-entropy:

$$\mathcal{L}_{\text{AU}}^{\text{VAS}} = \frac{1}{M} \sum_{m=1}^M \text{BCE}(y_{\text{AU}}^{(m)}, \phi_{\text{AU}}^{(m)}(\mathbf{f}_{\text{AU}}^{(m)})). \quad (3)$$

2.2. Self-organized Archetype Discovery (SAD) Module

This module serves as AURA’s geometric core, populating the continuous projection space with discrete, semantically rich visual archetypes. Unlike rigid clustering methods with fixed representational capacity, it adaptively captures the intrinsic data topology by assigning higher archetype density to complex affective states while preserving sparsity for simpler ones. This self-organization is achieved through two mechanisms: **Discriminative Archetype Assignment** anchors visual features to distinct semantic extremes to ensure representational fidelity, while **Adaptive Archetype Regularization** regularizes the archetype topology to promote density allocation according to affective complexity.

2.2.1. DISCRIMINATIVE ARCHETYPE ASSIGNMENT

This stage anchors each projected VAS feature to its highest-affinity archetype on the learned manifold. Let $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\} \subset \mathbb{R}^d$ denote a learnable set of K archetypes. Given a VAS feature $\mathbf{f} \in \mathbb{R}^d$, we first normalize $\bar{\mathbf{f}} = \mathbf{f}/\|\mathbf{f}\|_2$ and $\bar{\mathbf{a}}^k = \mathbf{a}^k/\|\mathbf{a}^k\|_2$. The assignment operator $\mathcal{Q}(\cdot; \mathcal{A})$ then selects the archetype with highest similarity:

$$\mathbf{a}^{k^*} = \mathcal{Q}(\mathbf{f}; \mathcal{A}), \quad k^* = \arg \max_k \langle \bar{\mathbf{f}}, \bar{\mathbf{a}}^k \rangle. \quad (4)$$

To ensure the archetypes are both *geometrically stable* and *semantically faithful*, we minimize two objectives:

(i) **Archetype Assignment Loss** ($\mathcal{L}^{\text{Assign}}$). To stabilize archetype discovery and feature discretization, we define a dual-term objective for each VAS feature $\mathbf{f}$ and its assigned archetype $\mathbf{a}^{k*}$. The first term adapts the archetype to the normalized VAS feature distribution, while the second imposes a commitment constraint on $\mathbf{f}$, preventing feature drift from the assigned archetype. Formally,

$$\mathcal{L}^{\text{Assign}} = \underbrace{\|\text{sg}[\bar{\mathbf{f}}] - \bar{\mathbf{a}}^{k*}\|_2^2}_{\text{archetype adaptation}} + \beta \underbrace{\|\bar{\mathbf{f}} - \text{sg}[\bar{\mathbf{a}}^{k*}]\|_2^2}_{\text{commitment penalty}}, \quad (5)$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operator and β balances archetype adaptation and feature commitment. This formulation allows archetypes to self-organize around the feature distribution while encouraging projected features to remain aligned with the discovered archetype structure.

(ii) **Archetype Discriminative Loss** ($\mathcal{L}^{\text{Dis}}$). We impose task-aware supervision on the assigned archetypes, allowing them to acquire task-aligned affective semantics. This provides the semantic basis for subsequent topology learning across FER, VA regression, and AU detection.

For FER. We define a shared archetype set $\mathcal{A}_{\text{Exp}} = \{\mathbf{a}_{\text{Exp}}^1, \dots, \mathbf{a}_{\text{Exp}}^{K_{\text{Exp}}}\}$ across all expression classes. Given a VAS feature $\mathbf{f}$, its assigned archetype is $\mathbf{a}_{\text{Exp}} = \mathcal{Q}(\mathbf{f}; \mathcal{A}_{\text{Exp}})$. The archetype prediction $\hat{\mathbf{y}}_{\mathbf{a}} = \phi_{\text{Exp}}(\mathbf{a}_{\text{Exp}})$ is optimized with the same loss as in Eq. (1), denoted as $\mathcal{L}_{\text{Exp}}^{\text{Dis}}$.

For VA Regression. Since valence and arousal describe distinct affective dimensions, we define two archetype sets, $\mathcal{A}_V$ and $\mathcal{A}_A$. VAS features $\mathbf{f}_V$ and $\mathbf{f}_A$ are assigned independently: $\mathbf{a}_V = \mathcal{Q}(\mathbf{f}_V; \mathcal{A}_V)$ and $\mathbf{a}_A = \mathcal{Q}(\mathbf{f}_A; \mathcal{A}_A)$. Their predictions $\hat{\mathbf{y}}_V = \phi_V(\mathbf{a}_V)$ and $\hat{\mathbf{y}}_A = \phi_A(\mathbf{a}_A)$ are supervised by the objective as in Eq. (2), denoted as $\mathcal{L}_{VA}^{\text{Dis}}$.

For AU Detection. To handle the multi-label nature of AUs, we define a distinct archetype set $\mathcal{A}_{\text{AU}}^{(m)}$ for each Action Unit m . Given a patch-level VAS feature $\mathbf{f}_{\text{AU}}^{(m)}$, its assigned archetype is $\mathbf{a}_{\text{AU}}^{(m)} = \mathcal{Q}(\mathbf{f}_{\text{AU}}^{(m)}; \mathcal{A}_{\text{AU}}^{(m)})$. The AU prediction $\hat{\mathbf{y}}_{\text{AU}}^{(m)} = \phi_{\text{AU}}^{(m)}(\mathbf{a}_{\text{AU}}^{(m)})$ is optimized with the same loss as in Eq. (3), denoted as $\mathcal{L}_{\text{AU}}^{\text{Dis},(m)}$.

2.2.2. ADAPTIVE ARCHETYPE REGULARIZATION

To ensure the learned archetype manifold captures the intrinsic geometry of affective states, AURA introduces an adaptive regularization mechanism for *Topology Refinement*. This mechanism shapes the topology through two complementary geometric criteria: *Density*, which controls representational capacity by assigning denser archetypes to complex, high-variance states and fewer to simpler ones, and *Expressiveness*, which preserves semantic fidelity by encouraging compactness within homogeneous regions and separation between distinct ones, with the refinement instantiated according to output space to preserve task semantics.

(i) Topology Refinement for Discrete Affect.

For discrete expression classification, topology refinement aims to partition the archetype space into semantically distinct regions while retaining intra-class visual diversity. We devise a strategy that enforces (a) intra-class compactness, (b) inter-class separation, and (c) global diversity. Through the interplay of these objectives, the archetype space self-organizes its density according to the visual complexity.

Formally, each archetype $\mathbf{a}_{\text{Exp}}^k$ is assigned to an expression class via $c^k = \arg \max_u [\phi_{\text{Exp}}(\mathbf{a}_{\text{Exp}}^k)]_u$, where $\phi_{\text{Exp}}(\cdot)$ is the classifier used in Eq. (1). Its *confidence* is measured by $\rho^k = \max_u [\text{softmax}(\phi_{\text{Exp}}(\mathbf{a}_{\text{Exp}}^k))]_u$. For each class c , we define the associated index set as $\mathcal{I}_c = \{k \mid c_k = c\}$ and compute its *class center* as $\boldsymbol{\mu}_c = \frac{1}{|\mathcal{I}_c|} \sum_{k \in \mathcal{I}_c} \bar{\mathbf{a}}_{\text{Exp}}^k$, with $\tilde{\boldsymbol{\mu}}_c = \boldsymbol{\mu}_c / \|\boldsymbol{\mu}_c\|_2$. The three optimization objectives are:

(a) *Intra-class Compactness* ($\mathcal{L}_{\text{intra}}$): This term anchors archetypes to their semantic centers to promote class coherence. Low-confidence archetypes receive larger penalties:

$$\mathcal{L}_{\text{intra}} = \frac{1}{K} \sum_{k=1}^K (1 - \rho^k) \cdot (1 - \cos(\bar{\mathbf{a}}_{\text{Exp}}^k, \tilde{\boldsymbol{\mu}}_{c_k})). \quad (6)$$

(b) *Inter-class Separation* ($\mathcal{L}_{\text{inter}}$): This term enforces distinctiveness between expression classes by penalizing center pairs whose similarity exceeds a margin γ . Let $\mathcal{S}_{\text{inter}} = \{(c, c') \mid c < c'\}$ denote the set of class-center pairs:

$$\mathcal{L}_{\text{inter}} = \frac{1}{|\mathcal{S}_{\text{inter}}|} \sum_{(c, c') \in \mathcal{S}_{\text{inter}}} \max(0, \cos(\tilde{\boldsymbol{\mu}}_c, \tilde{\boldsymbol{\mu}}_{c'}) - \gamma). \quad (7)$$

(c) *Archetype Diversity Loss* ($\mathcal{L}_{\text{div}}$): This term encourages semantic dispersion while preventing collapse by balancing global spread and class-wise distance variance. Let $\bar{\boldsymbol{\mu}} = \frac{1}{C} \sum_{c=1}^C \tilde{\boldsymbol{\mu}}_c$ be the global mean, $d_c = \|\tilde{\boldsymbol{\mu}}_c - \bar{\boldsymbol{\mu}}\|_2$, and $\bar{d} = \frac{1}{C} \sum_{c=1}^C d_c$. The loss is:

$$\mathcal{L}_{\text{div}} = -\frac{1}{C} \sum_{c=1}^C d_c + \frac{1}{C} \sum_{c=1}^C (d_c - \bar{d})^2. \quad (8)$$

The discrete topology loss is $\mathcal{L}_{\text{Exp}}^{\text{Top}} = \mathcal{L}_{\text{intra}} + \mathcal{L}_{\text{inter}} + \mathcal{L}_{\text{div}}$.

(ii) Topology Refinement for Continuous Manifolds.

Unlike FER, VA regression and AU detection are not naturally organized as mutually exclusive expression classes. VA lies in a continuous affective space. Although AUs are often treated as binary labels, their activations can vary in intensity (e.g., from “trace” to “strong” activation), making a smooth manifold representation more appropriate. To approximate this continuous landscape, we introduce a *score-aware learning strategy*, in which spatial proximity among archetypes reflects similarity in their predicted task scores.

(a) *Score-aware Attraction* ($\mathcal{L}_{\text{attr}}$): This term promotes manifold continuity by attracting archetypes with similar predicted affective estimates. Let $\hat{\mathbf{y}}_{\tau}^p = \phi_{\tau}(\mathbf{a}_{\tau}^p)$ and

$\hat{y}_\tau^q = \phi_\tau(\mathbf{a}_\tau^q)$ denote the affective estimates of archetypes $\mathbf{a}_\tau^p$ and $\mathbf{a}_\tau^q$, and let $d^{pq} = 1 - \cos(\bar{\mathbf{a}}_\tau^p, \bar{\mathbf{a}}_\tau^q)$ be their cosine distance. We define $\mathcal{S}_{\text{attr}} = \{(p, q) \mid p < q, |\hat{y}_\tau^p - \hat{y}_\tau^q| < \epsilon\}$, where ϵ sets score similarity and δ the distance margin, and penalize pairs with $d_{pq} > \delta$:

$$\mathcal{L}_{\text{attr}} = \frac{1}{|\mathcal{S}_{\text{attr}}|} \sum_{(p,q) \in \mathcal{S}_{\text{attr}}} [\max(0, d^{pq} - \delta)]^2. \quad (9)$$

(b) *Score-aware Repulsion* ($\mathcal{L}_{\text{repu}}$): This term expands the manifold by separating archetypes with distinct affective estimates. We define $\mathcal{S}_{\text{repu}} = \{(p, q) \mid p < q, |\hat{y}_\tau^p - \hat{y}_\tau^q| \geq \epsilon\}$ and penalize pairs whose distance falls below margin δ :

$$\mathcal{L}_{\text{repu}} = \frac{1}{|\mathcal{S}_{\text{repu}}|} \sum_{(p,q) \in \mathcal{S}_{\text{repu}}} [\max(0, \delta - d^{pq})]^2. \quad (10)$$

(c) *Archetype Diversity* ($\mathcal{L}_{\text{div}}$): For continuous settings, we compute the diversity regularization in the same way as Eq. (8), but over the normalized archetypes within each task-specific archetype space. This term discourages archetypes from concentrating in a limited region of the manifold and promotes broader semantic coverage.

The continuous topology loss is $\mathcal{L}_{\text{cont}}^{\text{Top}} = \mathcal{L}_{\text{attr}} + \mathcal{L}_{\text{repu}} + \mathcal{L}_{\text{div}}$.

2.2.3. TOTAL SAD LOSS

We combine the assignment, discriminative, and topology objectives. The total SAD loss is defined as:

$$\mathcal{L}^{\text{SAD}} = \mathcal{L}^{\text{Assign}} + \mathcal{L}^{\text{Dis}} + \mathcal{L}^{\text{Top}}, \quad (11)$$

where $\mathcal{L}^{\text{Dis}}$ and $\mathcal{L}^{\text{Top}}$ follow task-specific definitions above.

2.3. Archetype Contextualization (AC) Module

The SAD module establishes the physical geometry of the archetype manifold by learning the placement, density, and topology of visual archetypes. Yet affective prediction depends on the semantic geometry induced around each input, where the selected archetype must be interpreted in relation to its neighboring archetypes. The *Archetype Contextualization* module therefore performs distance-aware neighborhood reasoning, transforming geometric assignment into a context-conditioned semantic representation that captures transitions, intensity variation, and ambiguity.

2.3.1. SEMANTIC TOKEN PROJECTOR

To enable context-aware interaction between the input and retrieved archetypes, the *Semantic Token Projector* distills frozen visual features into compact semantic tokens for subsequent contextualization. Following the VAS projection strategy (Sec. 2.1), we use task-specific designs for global tasks (FER/VA) and local AU detection, ensuring that the tokens capture the appropriate semantic granularity.

(i) **Global Semantic Token Projector.** Given the global visual feature $\mathbf{z}_g \in \mathbb{R}^D$, we use learnable semantic queries to extract task-relevant affective attributes from the image-level representation. Let $\mathbf{Q} = [\mathbf{q}^1; \dots; \mathbf{q}^{N_{\text{tok}}}] \in \mathbb{R}^{N_{\text{tok}} \times d_{\text{tok}}}$ denote the semantic query tokens, where each query is encouraged to capture a latent affective attribute (e.g., overall intensity, valence polarity). The global feature is projected into key and value context tokens by $\mathcal{F}_K$ and $\mathcal{F}_V$, yielding $\mathbf{K}, \mathbf{V} \in \mathbb{R}^{N_{\text{ctx}} \times d_{\text{tok}}}$, where N_{ctx} denotes the number of context tokens. The global semantic token set is obtained through a multi-head cross-attention operator $\Psi(\cdot)$:

$$\mathbf{T}_g = \Psi(\mathbf{Q}, \mathbf{K}, \mathbf{V}) \in \mathbb{R}^{N_{\text{tok}} \times d_{\text{tok}}}. \quad (12)$$

(ii) **Patch Semantic Token Projector.** We use the patch embeddings $\mathbf{z}_p \in \mathbb{R}^{L \times D}$ to construct patch-aware semantic tokens. Specifically, we prepend N_{tok} learnable tokens $\{\mathbf{s}^n\}_{n=1}^{N_{\text{tok}}}$, with $\mathbf{s}^n \in \mathbb{R}^D$, to the patch sequence and process the concatenated tokens with a Transformer $\mathcal{T}_{\text{tok}}(\cdot)$:

$$[\tilde{\mathbf{t}}^1; \dots; \tilde{\mathbf{t}}^{N_{\text{tok}}}] = \mathcal{T}_{\text{tok}}([\mathbf{s}^1; \dots; \mathbf{s}^{N_{\text{tok}}}] \parallel \mathbf{z}_p), \quad \tilde{\mathbf{t}}^n \in \mathbb{R}^D.$$

For each AU m , the token outputs are projected through an AU-specific linear layer to obtain $\mathbf{t}^{(m),n} \in \mathbb{R}^{d_{\text{tok}}}$, forming the patch-aware semantic token set $\mathbf{T}_p^{(m)} \in \mathbb{R}^{N_{\text{tok}} \times d_{\text{tok}}}$.

2.3.2. DISTANCE-AWARE CONTEXTUALIZATION

To formulate the local archetype manifold as a structured semantic context, we introduce *Distance-aware Contextualization*. An input’s affective meaning is shaped by how it is positioned within its activated archetypal neighborhood, where both the retrieved archetypes and their geometric distances to the sample reveal semantic transitions and ambiguity. We therefore jointly encode neighboring archetypes and their sample-specific distances, enabling context-conditioned refinement of the local archetype structure.

We retrieve the top- R nearest archetypes $\{\mathbf{a}^{(1)}, \dots, \mathbf{a}^{(R)}\}$ for the sample feature $\mathbf{f}$ and compute their cosine distances as $d^{(r)} = 1 - \cos(\bar{\mathbf{f}}, \bar{\mathbf{a}}^{(r)})$. Each retrieved archetype is then converted into a *distance-aware* token by jointly encoding its semantic content and sample-specific distance:

$$\hat{\mathbf{a}}^{(r)} = \mathbf{W}_{\text{proj}} \mathbf{a}^{(r)} + \mathbf{b}_{\text{proj}} + \mathbf{W}_{\text{dist}} d^{(r)} + \mathbf{b}_{\text{dist}}, \quad (13)$$

where $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{d_{\text{tok}} \times d}$, $\mathbf{W}_{\text{dist}} \in \mathbb{R}^{d_{\text{tok}} \times 1}$, and $\mathbf{b}_{\text{proj}}, \mathbf{b}_{\text{dist}} \in \mathbb{R}^{d_{\text{tok}}}$ are learnable parameters. The sequence $\hat{\mathbf{A}} = [\hat{\mathbf{a}}^{(1)}; \dots; \hat{\mathbf{a}}^{(R)}] \in \mathbb{R}^{R \times d_{\text{tok}}}$ forms the retrieved archetypal context for local semantic structure refinement.

To model the semantic interaction between the input representation and its retrieved archetypal context, we assemble a composite sequence from the task-specific semantic token set $\mathbf{T}$ defined in Sec. 2.3.1. Two learnable $[\text{CLS}]$ tokens, $\mathbf{c}_{\text{sem}}$ and $\mathbf{c}_{\text{arc}}$, are inserted to summarize the *semantic-token stream* and the *archetypal-context stream*, respectively:

$$\mathbf{X}_{\text{ctx}} = [\mathbf{c}_{\text{sem}}; \mathbf{T}; \mathbf{c}_{\text{arc}}; \hat{\mathbf{A}}] \in \mathbb{R}^{(2+N_{\text{tok}}+R) \times d_{\text{tok}}}. \quad (14)$$

The sequence is processed by a Transformer encoder $\mathcal{T}_{\text{ctx}}(\cdot)$, yielding two contextual summaries $\tilde{\mathbf{c}}_{\text{sem}}$ and $\tilde{\mathbf{c}}_{\text{arc}}$. A gating module $\mathcal{G}(\cdot)$ estimates fusion weights $\alpha = [\alpha_{\text{sem}}, \alpha_{\text{arc}}]$ from these summaries, producing $\mathbf{f}_{\text{fused}} = \alpha_{\text{sem}}\tilde{\mathbf{c}}_{\text{sem}} + \alpha_{\text{arc}}\tilde{\mathbf{c}}_{\text{arc}}$. The fused representation is then passed to the task head $\phi_{\tau}(\cdot)$ to obtain the contextualized prediction $\hat{y}_{\tau}^{\text{ctx}}$.

This module is supervised by $\mathcal{L}^{\text{CTX}} = \mathcal{L}_{\text{task}}(\hat{y}_{\tau}^{\text{ctx}}, y)$, where $\mathcal{L}_{\text{task}}$ denotes the task-specific objective, i.e., CE for FER, BCE for AU detection, and the CCC-MSE objective for VA regression. This supervision encourages the contextualized archetype neighborhood to form a coherent semantic structure, supporting fine-grained affective inference.

3. Training Objective and Inference Strategy

Adaptive Archetype Self-Organization. A key outcome of AURA is the emergence of adaptive representational capacity. Unlike methods that prescribe the number of prototypes *a priori*, AURA initializes an over-complete archetype pool with an upper bound K_{max} to ensure sufficient capacity for diverse affective patterns. Driven by the topology and contextualization objectives, redundant archetypes receive limited activation, while semantically informative ones are repeatedly selected, refined, and integrated into the learned manifold. As training stabilizes, the model converges to a compact and stable active set $K_{\text{stable}} \ll K_{\text{max}}$ (see Appendix F.2 for stability analysis). This allows the complexity of the learned manifold to be governed by the data distribution rather than by a manually fixed prototype count. The detailed reset procedure is described in Appendix B.

Training Objective. The learning process is governed by a composite objective that jointly optimizes visual projection, archetype self-organization, and distance-aware contextualization. The total loss is defined as:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{VAS}}\mathcal{L}^{\text{VAS}} + \lambda_{\text{SAD}}\mathcal{L}^{\text{SAD}} + \lambda_{\text{CTX}}\mathcal{L}^{\text{CTX}}, \quad (15)$$

where λ_{VAS} , λ_{SAD} , and λ_{CTX} are balancing coefficients.

Inference Strategy: Perceptual Read-out. At inference time, we fix the discovered active archetypes and switch from contextual training to lightweight perceptual read-out. The contextualization module is omitted, since its training objective has already shaped the archetype space with local semantic structure. Given a visual embedding $\mathbf{z}$ for task τ , we project it into the archetype space as $\mathbf{f}_{\tau} = \mathcal{F}_{\tau}(\mathbf{z})$ and retrieve the nearest semantic archetype $\mathbf{a}_{\tau}^* = \mathcal{Q}(\mathbf{f}_{\tau}; \mathcal{A}_{\tau})$ by cosine similarity. The final prediction is obtained by applying the task head to the retrieved archetype, $\hat{y}_{\tau} = \phi_{\tau}(\mathbf{a}_{\tau}^*)$. This efficient pipeline relies *solely* on the projection module and visual archetypes, operationalizing direct perceptual resemblance with negligible computational overhead.

4. Experiments

We evaluate AURA on six benchmark settings: FER on RAF-DB (Li et al., 2017) and AffectNet-7/8 (Mollahosseini et al., 2017); AU detection on EmotioNet (Fabian Benitez-Quiroz et al., 2016) and DISFA (Mavadati et al., 2013); and VA regression on AffectNet-VA (Mollahosseini et al., 2017). Implementation details are provided in Appendix A. Additional cross-dataset generalization and end-to-end efficiency analyses are reported in Appendices F.7 and F.8.

4.1. Comparison with SOTA & CLIP Variants

Comparison with SOTA. As summarized in Table 1, AURA achieves strong performance across all three affective tasks. In FER, AURA obtains SOTA results. In AU detection, it achieves the highest F1, outperforming prior methods including temporal-modeling approaches such as MDHR (Wang et al., 2024), FUXI (Zhang et al., 2023a), and AUNet (Romero et al., 2022). In VA regression, AURA attains the highest CCC, demonstrating AURA can effectively capture continuous affective structure.

Comparison with CLIP Variants. Under identical protocols, AURA consistently outperforms text-dependent baselines. Prompt-learning methods (CoOp (Zhou et al., 2022b), CoCoOp (Zhou et al., 2022a)) show clear limitations on continuous (VA) and fine-grained (AU) tasks, where subtle visual cues and label imbalance make fixed textual supervision less effective. While CLIPER (Li et al., 2024a) remains competitive in FER, it is not designed for VA regression and requires task-specific adaptation for AU detection, while introducing additional inference cost.

Training Efficiency. AURA achieves strong performance with low computational cost among comparison methods, as shown in Table 1. By relying on a frozen visual encoder and lightweight trainable modules, it avoids expensive backbone fine-tuning. At inference, the pipeline reduces to cosine similarity matching against learned archetypes, introducing negligible additional overhead. End-to-end efficiency is provided in Appendix F.8.

4.2. Geometric and Semantic Interpretation

AURA reconstructs an interpretable affective geometry that adapts to task-specific semantics (fine-grained archetype visualizations, feature-space comparisons, detailed VA/AU analyses, and human-grounded interpretability studies are provided in Appendices C.1, C.3, C.4, C.5, and C.7). Here we analyze AURA through four complementary lenses:

Manifold Reconstruction (VA Space). In the continuous valence-arousal space (Fig. 2, top right), archetypes self-organize into an ordered topology. *Arousal* archetypes densely populate the high-variance moderate range

Table 1. Comparison of AURA with SOTA methods. Results are reported in F1-score (%) for AU detection, accuracy for FER, and CCC for VA regression.

Facial Expression Recognition (Accuracy in %)					
Model	RAF-DB ↑	AffNet-7 ↑	AffNet-8 ↑	Params ↓	FLOPs ↓
MRAN (Chen et al., 2023)	90.03	66.31	62.48	60.52	3.89
AMP-Net (Liu et al., 2022)	89.25	64.54	61.74	105.67	4.73
MA-Net (Zhao et al., 2021)	88.42	64.53	60.29	63.50	3.65
DACL (Farzaneh & Qi, 2021)	87.78	65.20	—	103.04	1.92
EAC (Zhang et al., 2022)	89.99	65.32	62.13	29.50	10.30
DR-FER (Li et al., 2023)	91.61	67.54	63.60	48.20	13.20
TransFER (Xue et al., 2021)	90.91	66.23	—	65.20	15.30
VTFF (Ma et al., 2021)	88.14	64.80	61.85	51.80	6.08
FRA (Gao & Patras, 2024)	90.76	65.85	62.55	24.00	—
S2D (Chen et al., 2024)	92.57	67.62	63.08	9.00	—
FER-Vamba (Ma et al., 2025b)	92.37	67.34	64.55	4.12	0.58
TriBAN (Kim & Choi, 2025)	91.43	66.05	62.49	67.38	4.81
POSTER++ (Mao et al., 2025)	92.21	67.49	63.77	43.70	8.40
UA-FER (Zhou et al., 2025)	92.59	66.95	62.42	28.74	11.27
MHAN (Wang et al., 2025)	92.57	67.74	65.08	4.27	0.55
CLIPER (Li et al., 2024a)	91.61	66.29	61.98	149.25	103.7
CLIP-FT (Radford et al., 2021)	89.14	64.76	60.21	149.25	21.8
CoOp (Zhou et al., 2022b)	84.65	61.25	57.76	85.71	4.26
CoCoOp (Zhou et al., 2022a)	85.89	63.14	59.66	86.80	4.35
AURA (Ours)	94.04 ± 0.17	68.43 ± 0.15	65.16 ± 0.24	3.30	0.26

Action Unit Detection (F1 Score in %)		
Methods	DISEA	EmotionNet
KSRL (Chang & Wang, 2022)	64.5	—
CTC (Zhou et al., 2023)	57.3	64.4
FBNet (Kollias et al., 2019)	62.0	54.0
AUNet (Romero et al., 2022)	59.7	64.6
MEGraphAU (Luo et al., 2022)	62.4	64.9
SITU (Liu et al., 2023)	62.9	64.2
CLEF (Zhang et al., 2023b)	64.8	—
FUXI (Zhang et al., 2023a)	63.3	65.4
MCM (Zhang et al., 2024)	64.3	—
EmoLA (Li et al., 2024b)	65.1	—
MDHR (Wang et al., 2024)	66.2	—
CLIPER (Li et al., 2024a)	54.9	56.4
CLIP-FT (Radford et al., 2021)	60.2	62.3
CoOp (Zhou et al., 2022b)	58.4	59.7
CoCoOp (Zhou et al., 2022a)	59.9	61.6
AURA (Ours)	66.9 ± 0.3	67.3 ± 0.4

Valence-Arousal Estimation (CCC in %)			
Model	CCC-V	CCC-A	CCC-VA
VA-StarGAN (Kollias & Zafeiriou, 2020)	61.0	48.0	54.5
FaceBehaviorNet (Kollias et al., 2021)	66.0	58.0	62.0
Emotion-GCN (Antoniadis et al., 2021)	76.7	64.9	70.8
EMOCA (Daněšček et al., 2022)	77.0	68.0	72.5
EmoNet (Toissoul et al., 2021)	73.0	65.0	69.0
BFsGP (Fu et al., 2025)	72.4	64.5	68.5
CAGE (Wagner et al., 2024)	71.6	64.2	67.9
Ig3D (Dong et al., 2024)	72.4	65.0	68.7
CMFR (He & Da, 2025)	77.0	69.0	73.0
CLIP-FT (Radford et al., 2021)	70.2	62.6	66.4
CoOp (Zhou et al., 2022b)	51.2	37.7	44.5
CoCoOp (Zhou et al., 2022a)	52.5	38.3	45.4
AURA (Ours)	78.0 ± 0.4	70.2 ± 0.1	74.1 ± 0.3

$[-0.3, 0.3]$, while *valence* archetypes form a smooth trajectory from negative to positive affect. This structure indicates that AURA approximates the continuous affective landscape with organized archetypes, supporting fine-grained VA regression through local manifold geometry.

Local Discriminability (AU Space). In the Action Unit space (Fig. 2, bottom right), the geometry shifts toward activation-oriented structure. For AU12, AURA separates a dense “inactive” region from a distinct “high-activation” region with a clear margin. This shift from the continuous VA manifold to localized AU activation shows that AURA can adapt its latent geometry to the *intrinsic structure* of each task.

Fine-grained Taxonomy Reconstruction. AURA goes beyond coarse predefined labels (e.g., 7 basic emotions) by revealing fine-grained, visually grounded substructures within each category. Instead of treating an expression label as a single semantic unit, AURA decomposes it into

distinct visual modes according to its intrinsic structure. As shown in Fig. 2 (left), the visually diverse *Happiness* category expands into 28 archetypes, whereas the more stereotyped *Fear* category is represented by only 6. This yields a self-organized affective lexicon that offers finer visual interpretability than static ground-truth labels.

Transparent Model Decision Trail. Building on the learned affective lexicon, AURA provides a transparent decision trail. As shown on RAF-DB (Fig. 2, left), the framework identifies **which** archetype anchors an input and **why** it is selected through geometric alignment. It further reveals **what** canonical visual pattern the archetype represents, with P1 capturing subtle closed-mouth smiles and P2 encoding high-intensity laughter, thereby showing **how** AURA decomposes coarse emotion labels into fine-grained visual and intensity patterns.

4.3. Datasets & Prediction Diagnostic Capabilities

A distinct advantage of AURA is its ability to use archetype-level semantic structure to *audit annotation quality* and *rationalize model decisions* through explicit visual evidence.

Dataset Diagnosis. AURA serves as a semantic auditor by comparing learned visual structure against dataset labels. As shown in Fig. 4(a) and detailed in Appendix C.2, some samples annotated as “Neutral” are anchored to “Sadness” archetypes due to subtle sorrow-related cues, such as knitted brows, that may be overlooked during annotation. This demonstrates AURA’s ability to flag potential annotation inconsistencies and reveal taxonomy ambiguities through visually grounded affective signals.

Understanding Failure Logic. Crucially, AURA makes misclassifications interpretable by exposing their underlying visual logic (Fig. 8, detailed in Appendix E). Rather than appearing as arbitrary errors, these cases often follow direct perceptual resemblance. For instance, “Happiness” samples misclassified as “Fear” align with the Fear P1 archetype due to a shared “scream-like” configuration. Such assignments indicate that some failures reflect visually plausible fine-grained groupings, helping users trace errors to specific local geometric alignments rather than opaque label confusion.

4.4. Ablation and Hyperparameter Robustness

Ablation Study. Our analysis confirms that each AURA component plays a critical role in the overall framework. As shown in Appendix F, Table 8, removing the VAS Projector, the Archetype Contextualization module, or the Topology Refinement objective degrades performance, indicating their importance for stable representation learning and task-aligned feature granularity.

Hyperparameter Sensitivity. We conduct a sensitivity

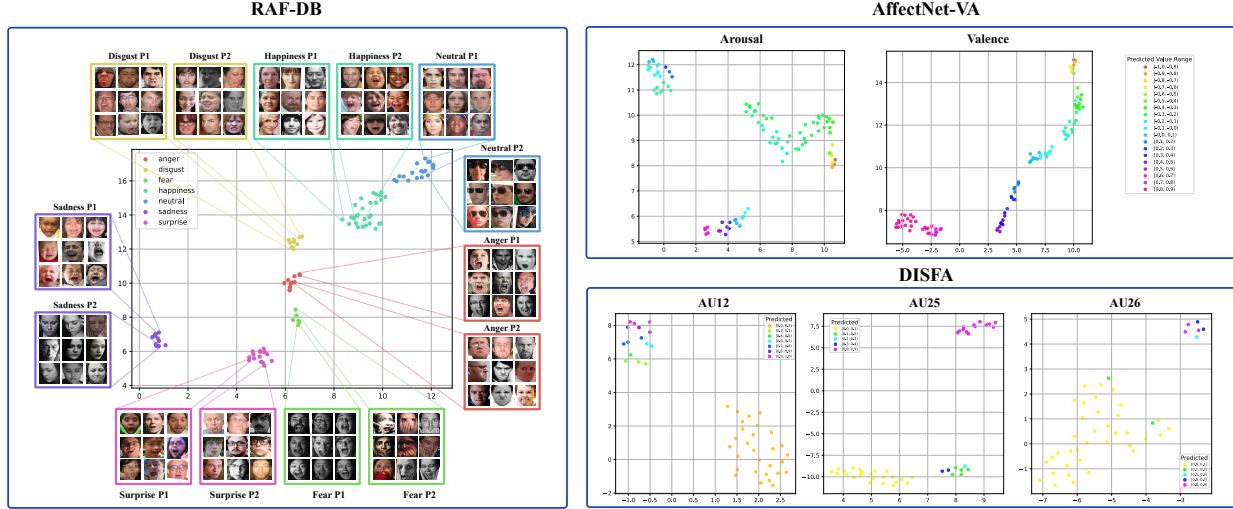


Figure 2. **Left:** visualization of learned archetype distributions on RAF-DB and their assigned samples, demonstrating the interpretability of AURA archetypes. **Right top:** archetypes on AffectNet-VA, illustrating fine-grained affective perception and their effectiveness for regression tasks. **Right bottom:** archetypes on DISFA for AU detection, showing clear distinction between active and inactive AUs.

analysis in Appendix F and observe that AURA is robust to hyperparameter choices. The loss weights λ_{VAS} , λ_{SAD} , λ_{CTX} achieve strong performance under the uniform setting $\lambda = 1$, as the objectives largely target different functional components. As a result, changing these weights mainly affects optimization dynamics, such as convergence speed, rather than the final accuracy (Table 10). Similarly, the distance margin exhibits a wide stable regime, with noticeable degradation only under extreme settings (Table 11).

4.5. Analysis of Archetype Adaptive Self-Organization

A key advance of AURA is *adaptive self-organization*, which learns state-adaptive archetype density. We verify this behavior through density statistics and sensitivity analysis.

Adaptive Density Allocation. Appendix Table 3 shows that archetype allocation is highly non-uniform and task-driven. **By complexity:** on RAF-DB, the visually diverse *Happiness* category is assigned 28 archetypes, whereas the less variable *Fear* category uses only 6. **By distribution:** on AffectNet-VA, AURA concentrates 34 archetypes in the dense central region $[-0.1, 0.1]$ and allocates only 1–5 to long-tail extremes, consistent with the data distribution. These results indicate that the AURA allocates representational capacity according to semantic complexity, avoiding concentration around a few dominant modes.

Robustness to Initialization ($K_{\max}$). $K_{\max}$ serves as an over-complete capacity upper bound, whereas the effective number of archetypes K_{stable} emerges through AURA’s self-organization and remains insensitive to the initial capacity. On RAF-DB with $K_{\max} \in \{150, 200, 300, 400\}$, K_{stable} stays tightly bounded between 99 and 102, while accuracy varies by at most 0.2%. This indicates that increasing $K_{\max}$ mainly increases computational cost without changing the

learned archetype granularity or representational structure.

Table 2. Evaluating with CLIP and DINO encoders. “FT” denotes full fine-tuning; “AURA” denotes frozen encoder + AURA.

Method	Encoder	RAF-DB Acc	AffectNet-VA CCC	EmotionNet AU-F1
CLIP FT	CLIP (finetuned)	89.1	66.4	62.3
AURA-CLIP	CLIP (official frozen)	94.0 (+4.9)	74.1 (+7.7)	67.3 (+5.0)
DINO FT	DINO (finetuned)	88.3	65.4	60.7
AURA-DINO	DINO (official frozen)	92.6 (+4.3)	72.0 (+6.6)	65.4 (+4.7)

4.6. Encoder-Agnostic Generalization

To verify that AURA’s performance stems from the *archetype mechanism* rather than solely from CLIP’s representational priors, we further evaluate it with the self-supervised DINO backbone (Caron et al., 2021). As shown in Table 2, AURA consistently improves over fully fine-tuned baselines for both CLIP and DINO. Notably, AURA-DINO with a frozen encoder outperforms DINO-FT by +4.3% on RAF-DB and +6.6 CCC points on AffectNet-VA. These results support the **backbone-agnostic** nature of AURA, showing that its mechanism can extend beyond vision-language alignment and enhance self-supervised visual representations through structured semantic anchors.

5. Conclusion

We proposed AURA, a *visually interpretable* archetype framework that moves beyond text-dependent VLM prompting through *archetype-grounded* geometric reasoning. AURA learns a self-organizing archetype manifold that adapts representational density to data complexity, supporting discrete and continuous affective prediction while providing transparent decision trails and dataset diagnostics. Across six benchmarks, AURA achieves SOTA performance with low overhead and encoder-agnostic generalization.

Impact Statement

This paper advances machine learning for affective computing by proposing a visually interpretable, archetype-based framework that can be instantiated across diverse affective tasks (expression classification, AU detection and valence–arousal regression). Like other affect recognition methods, it could be misused for privacy-invasive or non-consensual profiling; thus, any deployment should follow appropriate consent, governance, and domain-specific validation. These risks follow known concerns in affective computing, and we discuss the scope, limitations, and responsible-use considerations of AURA in Appendix G.

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A. Experiments

A.1. Datasets

AffectNet-7/8 & AffectNet-VA (Mollahosseini et al., 2017): AffectNet is an in-the-wild database that contains around 400K images manually annotated for 6 basic expressions, as well as neutral and contempt. For our work, we utilize the manually annotated images with the 7/8 expressions category to ensure alignment with other expression datasets. AffectNet-VA provides VA annotations in the range of $[-1, 1]$, making it suitable for dimensional affect analysis. The training set of this database consists of around 321K images and the validation of 5K. The validation set is balanced across the different expression categories. **RAF-DB (Real-world Affective Faces Database)** (Li et al., 2017): RAF-DB is an in-the-wild database that contains approximately 15,000 facial images, manually annotated for 7 basic expressions. **DISFA (Denver Intensity of Spontaneous Facial Action)** (Mavadati et al., 2013): DISFA is a lab controlled database consisting of videos from 27 subjects, each with approximately 5000 frames. Each frame is annotated with AU intensities on a six-point discrete scale (0–5). For consistency in AU detection tasks, we binarize the annotations, assigning a value of 1 to AU intensities greater than 2 and a value of 0 otherwise. The dataset includes annotations for 8 AUs (1, 2, 4, 6, 9, 12, 25, 26). **EmotionNet**(Fabian Benitez-Quiroz et al., 2016) consists of over 45K in-the-wild facial images, where we follow the official split and use the 11 most frequent AUs for training and evaluation.

A.2. Implementation Details

AURA is implemented in PyTorch and trained on a single NVIDIA A100 GPU. For all datasets, input images or video frames are first cropped to facial regions and resized to the input resolution required by the pretrained CLIP visual encoder. The CLIP visual encoder is kept frozen throughout training. We extract visual features once before optimization and then train AURA purely at the feature level, avoiding repeated forward passes through the large visual encoder.

Following the task-specific design in Sec. 2.1, we use global-level CLIP embeddings for FER and VA regression, and patch-level embeddings for AU detection. The global feature is taken from the image-level representation of the frozen encoder, while the patch feature sequence is taken from the final encoder layer and processed by the lightweight patch projector described in the main method. Unless otherwise specified, the default VAS dimension is set to $d = 32$, which provides the best overall trade-off in the ablation study.

All trainable modules in AURA, including the VAS projector, the Self-organized Archetype Discovery module, and the Archetype Contextualization module, are optimized with AdamW using a learning rate of 1×10^{-4} . A dropout rate of 0.2 is applied to the global-level and patch-level visual projectors. The loss weights are fixed across all datasets as $\lambda_{VAS} = \lambda_{SAD} = \lambda_{CTX} = 1$, without dataset-specific tuning. The commitment weight in the archetype assignment loss is set to $\beta = 1$. For the efficiency results, parameters and FLOPs are reported for the trainable AURA modules and feature-level inference after frozen feature extraction. The frozen CLIP encoder is not updated during AURA training. This follows the actual deployment pipeline of AURA, where features can be extracted once and predictions are made through the lightweight projector, archetype matching, and task heads. For encoder-agnostic experiments, the same protocol is applied after replacing CLIP with DINO, with the encoder kept frozen in the AURA-DINO setting.

A.3. Evaluation Protocols

Following prior affective understanding methods (Zheng et al., 2023; Jacob & Stenger, 2021; Luo et al., 2022; Hu et al., 2024; 2025d;a; Dong et al., 2024), we adopt task-specific evaluation metrics to ensure fair and meaningful performance comparisons across heterogeneous affective tasks.

Facial Expression Recognition. For FER, we report the classification accuracy (ACC), defined as:

$$\text{ACC} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\hat{y}_i = y_i), \quad (16)$$

where N is total number of samples, y_i is the ground-truth label, $\hat{y}_i$ is predicted label, and $\mathbb{I}(\cdot)$ is the indicator function.

Valence-Arousal Estimation. For VA estimation, we use the Concordance Correlation Coefficient (CCC) for both valence (v) and arousal (a), defined as:

$$\text{CCC}(x, y) = \frac{2\rho_{xy}\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2 + (\mu_x - \mu_y)^2}, \quad (17)$$

where ρ_{xy} is the Pearson correlation coefficient between the predicted values x and ground truth y , μ_x and μ_y are the means, and σ_x and σ_y are the standard deviations. The final CCC score is computed as the average of CCC_v and CCC_a :

$$\text{CCC}_{\text{VA}} = \frac{\text{CCC}_v + \text{CCC}_a}{2}. \quad (18)$$

Action Unit Detection. For AU Detection, we compute the F1-score for each Action Unit independently:

$$\text{F1}_k = \frac{2 \cdot \text{Precision}_k \cdot \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}, \quad (19)$$

where $\text{Precision}_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k}$ and $\text{Recall}_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k}$ for AU k . We further report the average F1-score across all K AUs:

$$\text{F1}_{\text{avg}} = \frac{1}{K} \sum_{k=1}^K \text{F1}_k. \quad (20)$$

B. Archetype Reset Mechanism

To avoid archetype collapse, we introduce a usage-aware reset mechanism that periodically reinitializes underutilized archetypes according to their selection frequency.

Global Usage Tracking: Let the over-complete archetype pool be denoted as $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^{K_{\max}}\}$, where each $\mathbf{a}^k \in \mathbb{R}^d$ is a learnable archetype. During each reset interval, we record the usage count vector $\mathbf{u} = [u_1, \dots, u_{K_{\max}}] \in \mathbb{N}_0^{K_{\max}}$, where u_k counts how often $\mathbf{a}^k$ is selected as the nearest archetype. The normalized usage ratio is defined as $\alpha_k = \frac{u_k}{\sum_{j=1}^{K_{\max}} u_j}$, $k = 1, \dots, K_{\max}$. Given a threshold $\eta_{\text{reset}} \in (0, 1)$, we identify the underutilized archetypes as $\mathcal{P}_{\text{reset}} = \{k \mid \alpha_k < \eta_{\text{reset}}\}$.

Archetype Reset: For each $k \in \mathcal{P}_{\text{reset}}$, we sample a VAS feature $\mathbf{f} \in \mathbb{R}^d$ from the current training batch and reinitialize the archetype as

$$\mathbf{a}^k \leftarrow \mathbf{f} + \boldsymbol{\xi}, \quad \boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}),$$

where $\sigma > 0$ denotes a small noise scale to encourage diversity. Alongside the archetype reset, we reinitialize its usage count and EMA statistics: $u_k \leftarrow 0$, $\mathbf{c}_k \leftarrow \mathbf{a}^k$, $n_k \leftarrow 1$, where $\mathbf{c}_k$ denotes the EMA feature accumulator for archetype k and n_k denotes its EMA assignment count. These statistics are used to update each archetype through the moving average of its assigned VAS features. After all underutilized archetypes are updated, we reset the global usage counter as $\mathbf{u} \leftarrow \mathbf{0}$. This mechanism helps the archetype pool adapt to the evolving feature distribution and prevents unused archetypes from becoming stale.

C. Analysis of Learned Archetypes

C.1. Fine-grained Archetype Structure Visualization

To provide a more direct view of the diversity, granularity, and semantic interpretability of the learned archetypes, we visualize the internal archetype structure of a representative expression category in Fig. 3. Using *Surprise* as an example, AURA discovers 15 active archetypes and arranges them into a structured topology rather than collapsing all samples into a single coarse representative. This visualization offers a concrete illustration of how AURA decomposes a broad affective category into multiple visually coherent submodes.

A key observation is that the learned archetypes are not distributed arbitrarily. Instead, they follow a meaningful facial organization along two dominant directions. The vertical direction roughly corresponds to *lower-face activation*, especially mouth opening, jaw drop, and lip-parting related configurations. The horizontal direction is more associated with *upper-face activation*, including eye widening, eyelid raising, brow-related cues, and gaze variation. Although this structure is not imposed manually, it emerges naturally from the learned archetype geometry, indicating that AURA captures regularities that align with interpretable facial semantics.



Figure 3. Fine-grained archetype structure visualized on a representative expression category. AURA allocates 15 active archetypes to *Surprise* and organizes them along an interpretable facial topology. The vertical direction roughly reflects lower-face activation, mainly mouth and jaw movement, while the horizontal direction captures upper-face activation, including eyes and eyelids. This topology reveals fine-grained variations within the category, showing that visually related but distinct facial patterns are preserved as separate archetypes.

This topology also reveals substantial fine-grained variation within the same expression category. Archetypes in the upper region are dominated by stronger mouth- and jaw-related activity, often accompanied by clear AU26-like jaw-drop patterns, which correspond to high-intensity surprised expressions. In contrast, archetypes in the lower region exhibit weaker lower-face activation and often preserve milder or more ambiguous surprised appearances. Along the horizontal direction, archetypes toward the right place greater emphasis on upper-face cues, such as widened eyes, bulging-eye configurations, eyelid activation, or mixed eye–mouth patterns, whereas those toward the left show comparatively weaker upper-face activation or alternative appearance factors such as side profile and side gaze. As a result, AURA separates different visual realizations of the same broad affective state while still preserving their higher-level semantic relatedness.

Another notable property is that neighboring archetypes remain semantically related but visually non-redundant. For example, several archetypes share strong lower-face activation but differ in whether the dominant complementary cue comes from eye widening, lip-parting, brow activity, or viewpoint variation. Likewise, archetypes with similar upper-face activation can still be distinguished by whether they co-occur with jaw drop, lip stretching, hand-over-mouth patterns, or profile changes. This shows that AURA does not merely split the category by intensity alone; instead, it preserves multiple interacting facial factors, yielding a richer and more faithful decomposition of intra-class structure.

Importantly, the visualization also highlights the semantic interpretability of individual archetypes. Each archetype is associated with a compact set of assigned samples that share a recognizable local facial pattern, and many of them can be linked to explicit facial cues such as AU26 (jaw drop), AU25 (lips part), AU5 (upper lid raiser), AU2 (outer brow raiser), or appearance patterns such as bulging eyes, side profile, and side gaze. This makes the learned structure directly inspectable: one can not only identify which archetype is activated, but also understand what canonical visual pattern that archetype represents.

Overall, Fig. 3 provides qualitative evidence that AURA learns a fine-grained and topologically organized archetype system. Rather than representing an expression category with a small set of coarse, isolated centers, AURA preserves intra-class diversity in a structured way, producing archetypes that are visually grounded, semantically interpretable, and organized according to meaningful facial variation. This visualization complements the quantitative results by showing how AURA turns broad categorical supervision into a high-resolution affective structure.

C.2. Diagnosing Annotation Ambiguities and Refining Emotion Taxonomy via AURA

Beyond representation analysis, AURA also provides a diagnostic view of dataset quality by comparing ground-truth labels with archetype assignments. Since each archetype is associated with a coherent set of support images, mismatches between the dataset label and the assigned archetype category can reveal potential annotation inconsistencies or ambiguous affective cases. This is particularly useful for large-scale in-the-wild datasets such as AffectNet, where compound expressions, low-intensity emotions, and subjective facial cues can make annotation difficult.

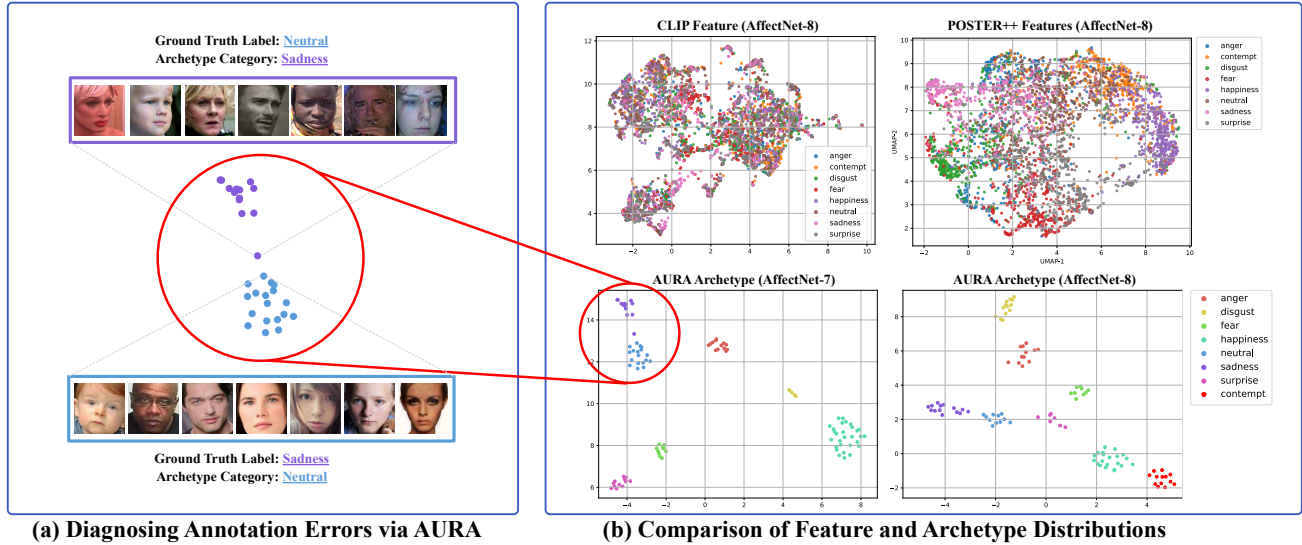


Figure 4. UMAP visualization of learned archetypes, original CLIP visual features, and POSTER++ features on AffectNet-7/-8. (a) AURA flags potential annotation inconsistencies by comparing ground-truth labels with archetype assignments. (b) Compared with CLIP and POSTER++ feature distributions, AURA forms compact and semantically separated archetype anchors.

As shown in Fig. 4(a), AURA identifies two closely related archetype groups associated with *Sadness* and *Neutral*. Some samples labeled as “Neutral” are assigned to *Sadness* archetypes. Visual inspection shows that these samples often contain sorrow-related cues, such as knitted brows, pronounced glabellar lines, drooping eyelids, dull gaze, or downward-compressed lips. Conversely, some samples labeled as “Sadness” are assigned to *Neutral* archetypes, and their support images tend to exhibit level eyebrows, relaxed eyelids, steady gaze, and lips at rest. These observations suggest that AURA can flag samples where the annotated label and the intrinsic visual structure are not fully aligned.

This diagnostic capability does not simply identify isolated labeling mistakes. It also reveals taxonomy ambiguity at the boundary between affective categories. For example, low-intensity sadness and neutral expressions can share similar facial configurations, making hard categorical labels insufficient to describe the underlying visual variation. By grouping samples according to archetype-level visual similarity, AURA exposes such borderline cases and provides a more structured explanation of why they are difficult to annotate.

More broadly, AURA refines the conventional emotion taxonomy into finer, visually grounded subsets. Instead of treating each emotion label as a single semantic unit, AURA decomposes it into multiple archetypal modes that capture intra-class variation, intensity differences, and compound affective patterns. This finer-grained organization helps distinguish mild from intense expressions and pure from mixed affective states. As a result, AURA offers a practical tool for analyzing dataset reliability, interpreting ambiguous samples, and improving the semantic granularity of affective taxonomies.

C.3. Affective Representation Advantage of AURA

AURA vs. Feature-based Representations. To assess how AURA organizes the affective space beyond conventional label-supervised representations, we visualize three types of representations on the AffectNet-8 test set (Fig. 4(b)): original CLIP features, POSTER++ features, and AURA archetypes. Original CLIP features are highly entangled across emotion categories, suggesting that generic vision-language representations alone provide limited affective separability. POSTER++ improves the class structure to some extent, but many samples remain intertwined, and the representation does not provide explicit semantic anchors for explaining the learned space.

In contrast, AURA does not merely reshape sample-level features. It learns a compact set of archetype anchors that organize the affective space into semantically separated regions. These anchors act as interpretable visual primitives, each corresponding to a coherent affective mode rather than an opaque feature cluster. As shown in the AURA archetype visualizations, the learned anchors exhibit clearer category-level structure than the raw CLIP and POSTER++ feature spaces, while preserving fine-grained intra-class organization. This indicates that AURA provides a visually grounded representation structure beyond conventional classification objectives, complementing its quantitative performance gains with explicit semantic interpretability.

C.4. Archetype Analysis in Valence-Arousal Space

AURA for Arousal. Fig. 5(left) shows that arousal archetypes form several localized regions in the UMAP space, with colors indicating their predicted arousal ranges. Rather than spreading uniformly, the archetypes exhibit denser coverage around moderate arousal values and sparser coverage near extreme low or high arousal states. This pattern is consistent with the empirical distribution of AffectNet-VA, where moderate arousal is more frequent and visually diverse, while extreme arousal states are less common and often visually more concentrated.

This organization suggests that AURA allocates representational capacity according to the density and variability of the continuous affective target. Moderate arousal covers a broad range of facial configurations, from calm attentiveness to mild agitation, and therefore benefits from a finer set of archetypes. In contrast, highly negative or highly positive arousal regions require fewer anchors, since they occupy smaller and less frequent parts of the data distribution. Thus, AURA does not simply discretize arousal into uniform bins. It learns a data-adaptive archetype topology that provides finer granularity where the affective manifold is more complex.

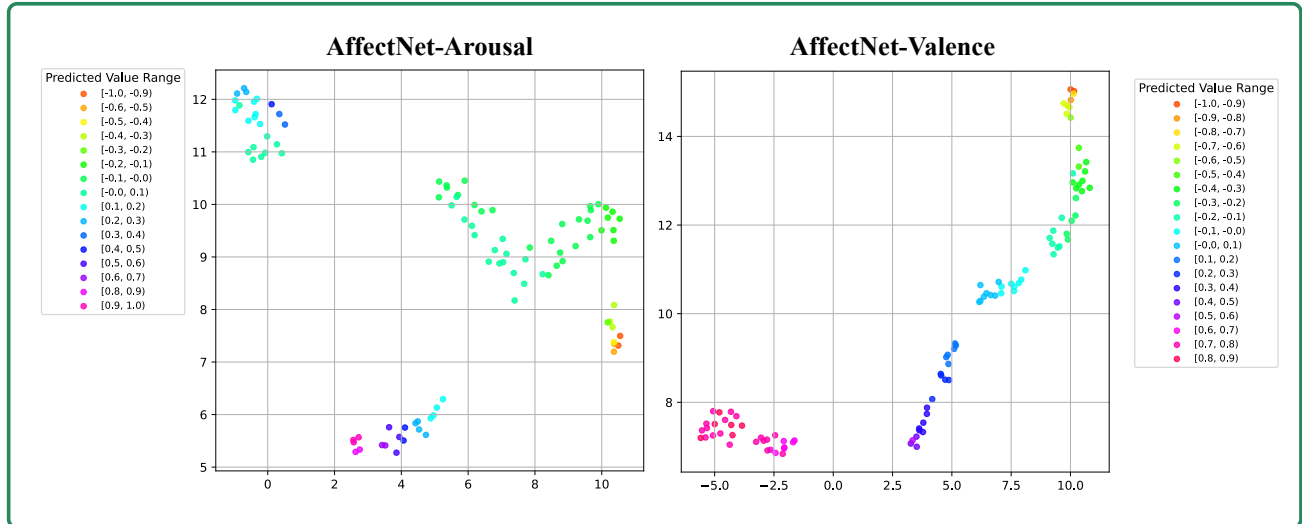


Figure 5. UMAP visualization of learned arousal and valence archetypes on AffectNet-VA. Colors indicate predicted value ranges. Arousal archetypes form several local regions with denser coverage around moderate affective intensities, while valence archetypes organize along an ordered trajectory from negative to positive affect.

AURA for Valence. The valence archetypes show a different structure. As shown in Fig. 5(right), they form an ordered trajectory that progresses from negative to positive affect, with nearby colors occupying neighboring regions. This trajectory-like organization indicates that AURA preserves the directional continuity of valence, where changes in archetype position

correspond to gradual shifts in affective polarity. Compared with arousal, whose archetypes form multiple local regions, valence exhibits a smoother and more monotonic semantic organization.

At the same time, the valence space is not uniformly populated. The visualization shows denser archetype allocation in visually important regions, especially around highly positive and mildly negative affective states, while still maintaining coverage across the full value range. This selective densification reflects the uneven distribution and visual complexity of valence in AffectNet-VA. Positive expressions often contain diverse smile-related patterns, while mildly negative states can include subtle or ambiguous facial cues. AURA captures these variations by allocating more archetypes to regions requiring finer semantic resolution.

Overall, the arousal and valence analyses show that AURA adapts its archetype geometry to the structure of continuous affective targets. Arousal is represented through localized density patterns that reflect intensity variability, while valence is represented through an ordered semantic trajectory from negative to positive affect. Together, these results indicate that AURA reconstructs the continuous VA manifold with both compactness and granularity, supporting interpretable and fine-grained regression.

C.5. Analysis of Archetypes in AU Spaces

We visualize the learned AURA archetypes for four representative Action Units: AU4, AU12, AU25, and AU26. As shown in Fig. 6, the archetype distributions are highly AU-specific. Rather than allocating archetypes uniformly across activation values, AURA organizes each AU space according to its intrinsic activation structure. In general, weak or inactive states occupy broader regions, reflecting larger appearance variability, while stronger activations tend to form more compact regions when their visual patterns are more distinctive.

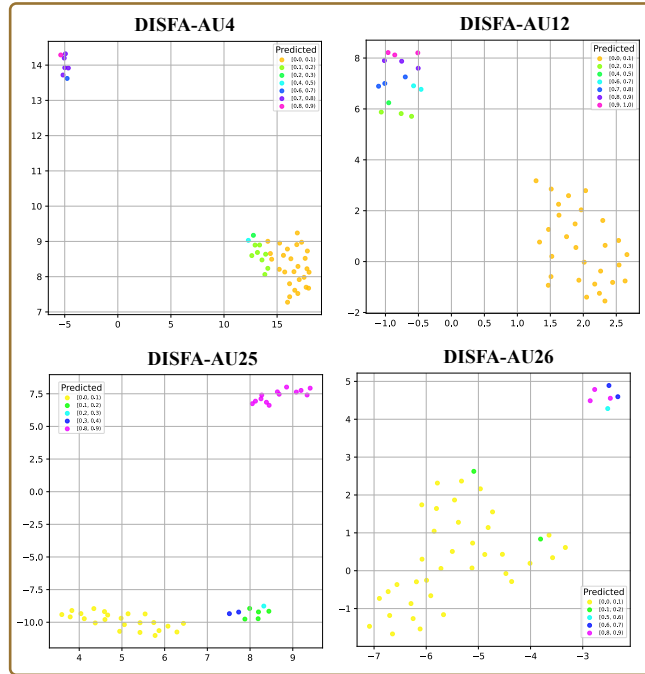


Figure 6. UMAP visualization of learned AURA archetypes for representative DISFA Action Units. Colors indicate predicted activation ranges. The distributions reveal AU-specific intensity structures, including inactive-active separation, graded activation transitions, and selective density allocation across different AU spaces.

For AU4, the archetypes form a clear two-region structure. Low-activation archetypes in the range $[0.0, 0.3]$ occupy a broad region on the right side of the UMAP space, while high-activation archetypes in the range $[0.6, 0.9]$ form a compact and well-separated group on the left side. The large gap between these two regions indicates that AURA separates inactive or weak brow-lowering patterns from strong AU4 activation, with few intermediate archetypes required.

For AU12, the structure is different. The low-activation archetypes, mainly in $[0.0, 0.1]$, form a large region on the right, whereas active archetypes from approximately $[0.2, 1.0]$ are grouped in a separate region on the left. Within this active region,

different activation levels remain locally organized, suggesting that AURA preserves the graded nature of smile-related activation rather than collapsing all active AU12 states into a single archetype. This is consistent with the continuous visual variation of AU12, where smile intensity can change smoothly across samples.

AU25 shows a three-level organization. Near-inactive archetypes in $[0.0, 0.1]$ occupy a broad lower region, intermediate activations around $[0.1, 0.4]$ form a smaller neighboring group, and high activations around $[0.8, 0.9]$ appear as a compact region clearly separated from the lower-intensity states. This indicates that AURA captures both coarse inactive-active separation and finer intensity-dependent structure within the AU25 space.

For AU26, low-activation archetypes in $[0.0, 0.1]$ are widely distributed along a broad trajectory, while stronger activations, mainly in $[0.5, 0.9]$, form a compact group in a separate region. This suggests that weak AU26 states are visually heterogeneous, whereas strong jaw-drop activations are more visually consistent and can be represented by fewer concentrated archetypes.

Overall, these visualizations show that AURA learns AU-specific archetype geometries that reflect the structure of facial muscle activation. The model separates inactive and active states when their visual difference is sharp, preserves graded transitions when activation changes continuously, and allocates more representational coverage to visually heterogeneous regions. This supports the role of AURA as a fine-grained archetype mechanism for AU detection, capturing both discriminative activation boundaries and subtle intensity variation.

C.6. Quantitative Analysis of Archetype Distribution Across Tasks

We provide a quantitative analysis of archetype allocation across three representative affective tasks: categorical facial expression recognition on RAF-DB, continuous arousal-valence estimation on AffectNet-VA, and AU detection on DISFA. Table 3 reports the number of active archetypes associated with each class or score bin, their usage ranges, the number of samples assigned to these archetypes, and the corresponding number of dataset samples under the ground-truth annotation. **Assigned Samples** and **Dataset Samples** reflect two different statistical views. Assigned Samples are counted according to the archetype selected by AURA, whereas Dataset Samples are counted according to the original ground-truth label or score bin. Therefore, the two numbers need not match exactly in each row. Their difference provides a useful view of how AURA reorganizes samples according to learned visual-semantic structure rather than strictly following annotation boundaries.

Table 3. Statistics of assigned archetypes across datasets and tasks. Each row reports the number of active archetypes associated with the corresponding class or score bin (**Assigned Archetypes**), the usage range of these archetypes (min–max, **Archetype Usage**), the number of samples assigned to these archetypes (**Assigned Samples**), and the number of dataset samples in the corresponding ground-truth class or bin (**Dataset Samples**).

RAF-DB				
Expression	Assigned Archetypes	Archetype Usage	Assigned Samples	Dataset Samples
anger	11	12–566	710	705
disgust	9	17–288	751	717
fear	6	18–169	287	281
happiness	28	19–1788	4735	4772
neutral	19	8–1288	2417	2524
sadness	12	11–999	2009	1982
surprise	15	14–841	1362	1290
AffectNet-VA (Arousal / Valence)				
Bin Range	Assigned Archetypes	Archetype Usage	Assigned Samples	Dataset Samples
−1.0 – −0.7	1 / 3	2341 / 1739–6783	2341 / 10522	3716 / 17989
−0.7 – −0.4	3 / 8	1013–5795 / 6741–7719	9206 / 36362	12372 / 31838
−0.4 – −0.1	28 / 13	1733–8666 / 1258–7383	63836 / 54243	52069 / 36753
−0.1 – 0.1	34 / 20	1389–7014 / 949–6734	71999 / 36927	99781 / 50891
0.1 – 0.4	22 / 17	1113–12569 / 154–3948	92903 / 20537	74212 / 29866
0.4 – 0.7	7 / 15	1178–3846 / 747–8513	21585 / 45156	30415 / 66280
0.7 – 1.0	5 / 24	5370–9079 / 1255–5565	28220 / 86343	21845 / 60793
DISFA (AU12 / AU25)				
Bin Range	Assigned Archetypes	Archetype Usage	Assigned Samples	Dataset Samples
0.0 – 0.3	32 / 29	471–5819 / 538–7391	62970 / 54047	65819 / 55054
0.3 – 0.5	1 / 2	1226 / 141–2440	1226 / 2581	
0.5 – 0.7	2 / 0	111–963 / 0	1074 / 0	
0.7 – 1.0	9 / 13	411–6869 / 769–4588	21940 / 30582	21391 / 32156

Note. For AffectNet-VA, values before and after “/” correspond to arousal and valence, respectively. For DISFA, values before and after “/” correspond to AU12 and AU25, respectively. In DISFA, **Assigned Samples** are grouped by the predicted activation ranges of archetypes, while **Dataset Samples** follow the ground-truth inactive/active groups used for AU detection.

RAF-DB (Expression Recognition): For categorical expression recognition, archetype allocation varies substantially across

the seven expression categories. Visually diverse and high-frequency categories receive more archetypes. For example, *happiness* is assigned 28 archetypes with a broad usage range of 19–1788, while *neutral* is assigned 19 archetypes with a usage range of 8–1288. This indicates that AURA allocates higher representational capacity to categories with larger intra-class variation and more heterogeneous facial patterns. In contrast, *fear* and *disgust* are represented by only 6 and 9 archetypes, respectively, suggesting more compact visual structures in the learned archetype space. *Anger*, *sadness*, and *surprise* fall in between, with moderate archetype counts and usage spans. These results show that AURA does not impose a fixed number of representatives per class, but adapts archetype density to the visual complexity of each expression category.

The difference between Assigned Samples and Dataset Samples further reveals the behavior of archetype-based grouping. For instance, the number of samples assigned to *anger* archetypes is 710, while the ground-truth number of anger samples is 705. Such differences indicate that some samples from other labels may be visually closer to anger archetypes, or that some anger-labeled samples may be assigned to neighboring affective modes. This supports the diagnostic role of AURA: archetypes expose visual-semantic structure that can differ from rigid ground-truth labels, especially for ambiguous or compound expressions.

AffectNet-VA (Arousal / Valence Estimation): For continuous affect estimation, AURA exhibits different allocation patterns for arousal and valence. In the arousal dimension, archetypes are concentrated around moderate intensity regions. The bins $[-0.4, -0.1]$, $[-0.1, 0.1]$, and $[0.1, 0.4]$ receive 28, 34, and 22 archetypes, respectively, while the extreme regions receive substantially fewer archetypes. This matches the empirical structure of AffectNet-VA, where moderate arousal states are more frequent and visually diverse, whereas extremely low or high arousal states are less common and often more visually concentrated. The high number of assigned samples in the moderate arousal bins further indicates that AURA allocates finer granularity where the continuous affective manifold is denser and more variable.

Valence follows a different pattern. Rather than concentrating only around the center, valence archetypes maintain broad coverage across the full $[-1, 1]$ range, with selective densification in important regions. In particular, the high-positive bin $[0.7, 1.0]$ receives 24 archetypes and 86,343 assigned samples, while the near-neutral bin $[-0.1, 0.1]$ receives 20 archetypes. This suggests that positive affective states, especially smile-related expressions, contain rich visual variation that requires multiple archetypes, while near-neutral states also require finer modeling due to subtle and ambiguous facial cues. Thus, AURA does not uniformly discretize the VA space. It learns dimension-specific archetype allocation, with arousal emphasizing dense moderate-intensity regions and valence preserving an ordered coverage from negative to positive affect with selective high-density areas.

DISFA (Action Unit Detection): For AU detection, AURA allocates archetypes according to activation-dependent visual variability. In AU12, the low-activation range $[0.0, 0.3]$ dominates with 32 archetypes and 62,970 assigned samples, reflecting the large appearance variability of weak or inactive smile-related patterns. The mid-intensity range $[0.3, 0.5]$ is covered by only one archetype, while the high-activation range $[0.7, 1.0]$ uses 9 archetypes but accounts for 21,940 assigned samples. This suggests that high AU12 activation is visually more compact, whereas low activation contains more heterogeneous facial appearances.

AU25 shows a related but distinct pattern. The low-activation range $[0.0, 0.3]$ is assigned 29 archetypes and 54,047 samples, while high activation $[0.7, 1.0]$ is represented by 13 archetypes and 30,582 assigned samples. Notably, no archetype is assigned to the $[0.5, 0.7]$ range, indicating that this intermediate region is either rarely occupied or not selected as a stable archetypal mode under the learned AU25 structure. These statistics show that AURA does not reduce AU detection to a simple inactive-active binary decision. Instead, it captures activation-dependent structure by assigning more archetypes to visually heterogeneous regions and fewer archetypes to compact or rarely occupied intensity ranges.

Overall, Table 3 demonstrates that AURA learns data-adaptive archetype allocation across heterogeneous affective tasks. Expression categories with greater intra-class diversity receive more archetypes, continuous VA dimensions exhibit score-dependent density allocation, and AU spaces organize archetypes according to activation intensity and visual variability. This behavior reflects the core advantage of AURA: representational capacity is governed by the learned affective structure of the data, rather than by a manually fixed prototype count or uniform discretization. As a result, AURA achieves a compact yet expressive archetype space that supports both interpretability and fine-grained affective modeling.

C.7. Quantitative and Human-grounded Interpretability

Beyond qualitative visualization, we further evaluate whether the learned archetypes correspond to structurally meaningful facial patterns. We conduct two complementary analyses on RAF-DB: a landmark-based structural consistency test and a

human-grounded within-class similarity study.

Landmark-based structural consistency. We first examine whether samples assigned to the same archetype share more consistent facial geometry than samples assigned to other archetypes within the same class. For each test sample, we compute its normalized facial landmark distance to two types of references: (i) the centroid of its assigned archetype within the same class, and (ii) the centroids of other archetypes from the same class. If the learned archetypes capture meaningful within-class facial structure, the distance to the assigned archetype should be smaller.

As shown in Table 4, this pattern holds consistently across all expression classes. On average, the normalized landmark distance to the assigned archetype is 3.4, compared with 4.6 to other same-class archetypes, yielding a relative reduction of 24.9%. The reduction is especially clear for Neutral, Fear, Sadness, and Surprise, indicating that AURA does not merely group samples by coarse emotion labels, but further organizes them into geometrically coherent submodes.

Table 4. Within-class landmark consistency on RAF-DB. **Assigned** denotes the normalized landmark distance to the centroid of the assigned archetype, while **Other** denotes the distance to other same-class archetype centroids. Lower distance indicates stronger structural consistency.

Class	Assigned	Other	Reduction (%)
Anger	3.6	4.5	19.6
Disgust	4.0	4.9	17.3
Fear	2.9	4.1	28.4
Happiness	2.9	3.8	22.6
Neutral	3.3	5.0	32.8
Sadness	3.6	4.9	26.8
Surprise	3.5	4.8	26.9
Average	3.4	4.6	24.9

Human-grounded within-class evaluation. We further conduct a human-grounded evaluation to test whether the learned archetypes align with human perception of facial-structure similarity. We use all active archetypes learned by AURA on RAF-DB and sample three query images per archetype. For each query, annotators are shown two same-class archetype references: the AURA-assigned archetype and a same-class distractor archetype. They are asked to select which reference is more similar to the query in facial structure. Each query is judged by five independent annotators.

The results are reported in Table 5. Across all classes, annotators prefer the AURA-assigned archetype in 76.9% of cases, compared with 23.1% for the distractor. The majority-vote agreement is 74.4%, suggesting that the preference is not driven by isolated annotator noise. These results provide human-grounded evidence that AURA’s archetypes capture recognizable within-class facial structure, rather than only coarse category-level information.

Table 5. Human evaluation of within-class archetype interpretability on RAF-DB. Annotators choose whether the AURA-assigned archetype or a same-class distractor is more similar to the query image in facial structure.

Class	Prefer AURA (%)	Distractor (%)	Majority Agree (%)
Anger	77.0	23.0	74.6
Disgust	73.6	26.4	71.2
Fear	72.8	27.2	70.4
Happiness	81.5	18.5	78.0
Neutral	76.3	23.7	74.8
Sadness	75.4	24.6	73.1
Surprise	79.1	20.9	76.4
Average	76.9	23.1	74.4

Together, the landmark-based and human-grounded evaluations support the interpretability of AURA from two perspectives. The landmark analysis shows that assigned archetypes are geometrically closer to their samples than other same-class archetypes, while the human study shows that this structural organization is also perceptible to annotators. This complements the qualitative visualizations and provides quantitative evidence that AURA learns fine-grained, structurally coherent, and human-recognizable archetypes.

D. AURA vs. Prototype/VQ-style Methods: From Flat Codebooks to Topology-shaped Archetypes

This section situates AURA within the broader family of representative-vector paradigms, including prototype learning, codebook learning, and vector quantization. These paradigms share a common operation: mapping continuous features to a finite set of learned representatives. Their key difference lies in what those representatives encode and how their geometry is formed. AURA uses nearest-archetype assignment within an affective archetype manifold, where the learned anchors are shaped by task-aware supervision, adaptive topology refinement, and distance-aware contextualization. Under this formulation, archetypes are not isolated code entries or class-level averages, but semantic extrema whose density and neighborhood relations reflect the visual complexity of affective states. We first compare AURA with prototype- and codebook-based affective methods, and then further isolate its design through a controlled replacement with vanilla vector quantization.

D.1. Comparison with Prototype and Codebook-style Methods

Prior prototype-style methods in facial expression analysis can be broadly grouped into two categories. The first directly **learns** a fixed set of representative vectors. For example, ProS (Di et al., 2024) uses a large fixed bank of facial prototypes, while POST (Guo et al., 2024) learns one prototype per class. The second constructs prototypes by **aggregating** pretrained embeddings, such as MPA-FER (Ma et al., 2025a), which forms class-level prototypes by averaging frozen CLIP embeddings within each category. In both cases, the learned units are closer to fixed representative centers than to a self-organized semantic manifold.

AURA differs from these methods in both representation and learning dynamics. Instead of forcing an affective category into a single class-level center or a fixed prototype bank, AURA preserves topology-aware structural variation. When one affective region contains multiple distinct but related facial patterns, AURA can maintain multiple archetypes with different local geometry. This is important for affective analysis, where a single expression label often contains large visual diversity, compound patterns, and intensity variation.

More specifically, AURA differs from prior prototype and codebook-style methods in four aspects. First, it learns semantically meaningful visual archetypes rather than blurred class averages. Second, it uses adaptive density allocation rather than a manually fixed number of centers per class. Third, it supports classification, detection, and continuous regression within the same archetype-learning principle. Fourth, it explicitly models relations among archetypes through topology refinement and contextualization, rather than treating the learned units as independent code vectors.

Table 6. Comparison between AURA and representative prototype/codebook-style methods on RAF-DB. The comparison highlights whether the learned units support topology-aware interpretability, adaptive density, and explicit relations among units.

Method	Topology-aware	Density	Explicit Relations	RAF-DB Acc. (%)
ProS (Di et al., 2024)	No	Fixed (1024)	No	91.04
POST (Guo et al., 2024)	No	Fixed (7)	No	90.59
MPA-FER (Ma et al., 2025a)	No	Fixed (7)	No	93.74
AURA (Ours)	Yes	Adaptive	Yes	94.04

As shown in Table 6, AURA is not only stronger in performance, but also differs in what its learned units represent. Fixed-prototype methods can provide representative centers, but they do not explicitly organize these centers into a task-aware topology. AURA instead learns archetypes whose density and relations are adjusted by the data distribution and affective structure.

D.2. Controlled Replacement with Vanilla VQ

To further isolate the distinction between AURA and vanilla vector quantization (VQ), we conduct a controlled replacement experiment in which the encoder, projector, task heads, and total code budget are kept unchanged, while only the archetype learning strategy is replaced. This setup isolates the effect of AURA’s topology-aware and context-aware archetype learning from the nearest-assignment mechanism itself.

Difference in construction pipeline. Vanilla VQ typically follows a simple pipeline: feature extraction, nearest-code assignment, codebook update with commitment regularization, and dead-code reset. AURA also uses nearest-archetype

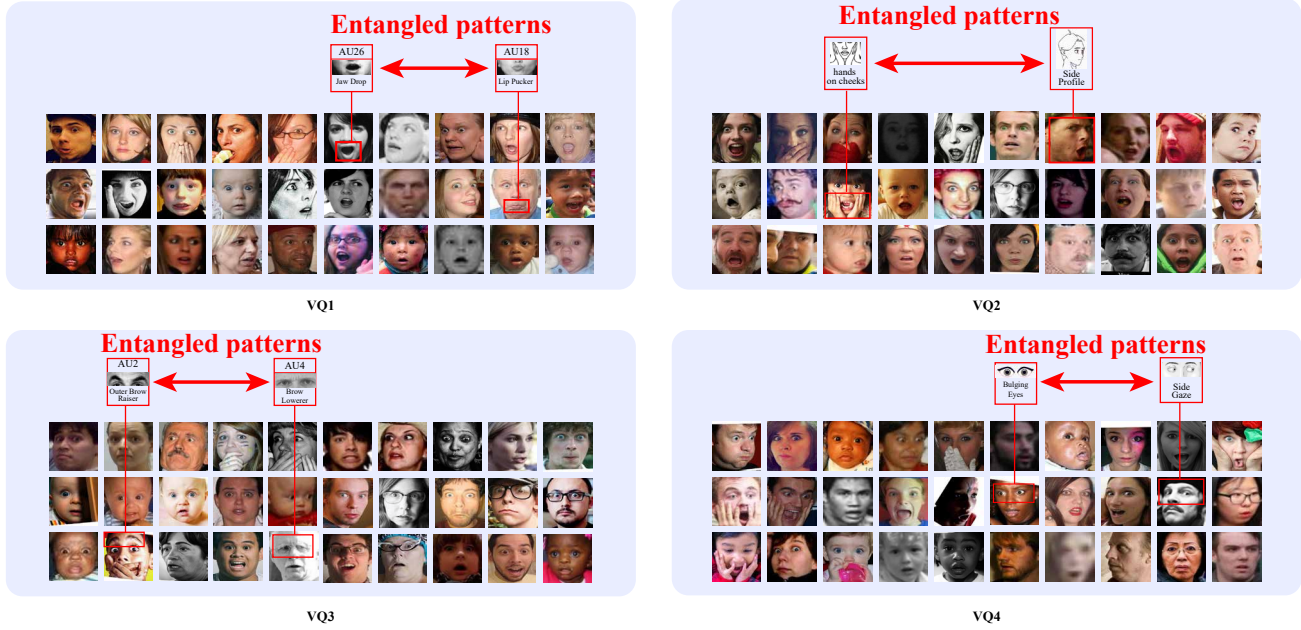


Figure 7. **Vanilla VQ codes for Surprise under the same code budget.** Unlike AURA, vanilla VQ activates only 4 codes for Surprise, and each code mixes structurally entangled facial patterns. Without topology shaping or contextualization, visually different configurations are forced into the same code, preventing the emergence of a fine-grained and interpretable facial topology.

assignment as a basic building block, but its contribution goes beyond assignment. After assignment, AURA further imposes topology refinement through regularization terms that encourage compactness, separation, and diversity, allowing the learned archetypes to reflect the geometry of the target task. In addition, AURA introduces contextualization between semantic tokens and retrieved neighboring archetypes, encouraging local archetype neighborhoods to form a coherent semantic manifold during training. Together with the usage-aware reset mechanism, these components yield an adaptively reorganized and non-uniform archetype distribution.

Table 7. Class-wise allocation of learned units and accuracy comparison between AURA and vanilla VQ under the same code budget on RAF-DB.

Method	anger	disgust	fear	happiness	neutral	sadness	surprise	Total	Acc. (%)
AURA	11	9	6	28	19	12	15	100	94.0
VQ	15	12	19	16	27	7	4	100	89.0

Difference in quantitative outcome. The quantitative comparison is summarized in Table 7. Under the same total budget of 100 learned units, AURA achieves substantially higher accuracy than vanilla VQ on RAF-DB, improving from 89.0% to 94.0%. The class-wise allocation also differs markedly. AURA distributes active archetypes in a way that better reflects affective complexity, assigning 28 archetypes to Happiness and 15 to Surprise. In contrast, vanilla VQ assigns only 4 codes to Surprise while allocating 19 and 27 codes to Fear and Neutral, respectively. This indicates that vanilla VQ does not reliably allocate representational capacity according to the fine-grained semantic complexity of each expression category.

The difference is also evident at the unit-usage level. In AURA, 93 archetypes each cover more than 50 samples, whereas in vanilla VQ only 47 codes exceed this threshold. This suggests that vanilla VQ tends to collapse many samples into a small number of dominant codes. A representative example is Sadness: among 2009 samples assigned to Sadness-related codes, one VQ code alone dominates the assignments with 1875 samples, while the remaining codes cover only 66, 37, 14, 9, 5, and 3 samples. Such extreme concentration indicates that vanilla VQ compresses broad intra-class variation into a few overloaded codes, whereas AURA maintains a richer and more distributed representation.

Difference in qualitative structure. The qualitative difference is illustrated in Figs. 3 and 7. For the Surprise category, AURA discovers 15 active archetypes organized into a structured facial topology. These archetypes vary systematically

along interpretable facial dimensions, especially lower-face activation and upper-face activation, allowing distinct but related surprise patterns to remain separate while still belonging to the same affective region.

Vanilla VQ, in contrast, activates only 4 codes for Surprise under the same overall budget. As shown in Fig. 7, each VQ code mixes structurally different facial patterns instead of isolating coherent visual modes. For example, one code entangles jaw drop with lip pucker, while another mixes side-profile cases with hands-on-cheeks patterns. Other codes combine outer-brow raising with upper-lid raising, or bulging eyes with side gaze, without forming a clear topology of facial variation. These entangled groupings indicate that, with only commitment-based optimization, VQ codes remain independent entries optimized mainly for assignment and prediction, and do not naturally acquire an interpretable semantic structure.

E. Failure Case Analysis

E.1. Archetypal Explanation of Misclassifications

To further examine the behavior of AURA, we conduct a failure-case analysis in Fig. 8. Each panel compares an expression archetype with misclassified samples that are assigned to it. The left block shows support samples that define the visual pattern of the archetype, while the right block shows samples whose ground-truth labels differ from the archetype category but are nevertheless matched to it. This setup allows us to inspect not only whether the model is wrong, but also which visual structure drives the error.

Across the panels, a consistent pattern emerges: many failures are visually interpretable rather than opaque label confusion. Misclassified samples often share local facial configurations with the assigned archetype, such as mouth opening, eye tension, brow contraction, or nose-mouth deformation. Thus, the assigned archetype provides a concrete visual explanation of the model’s decision.

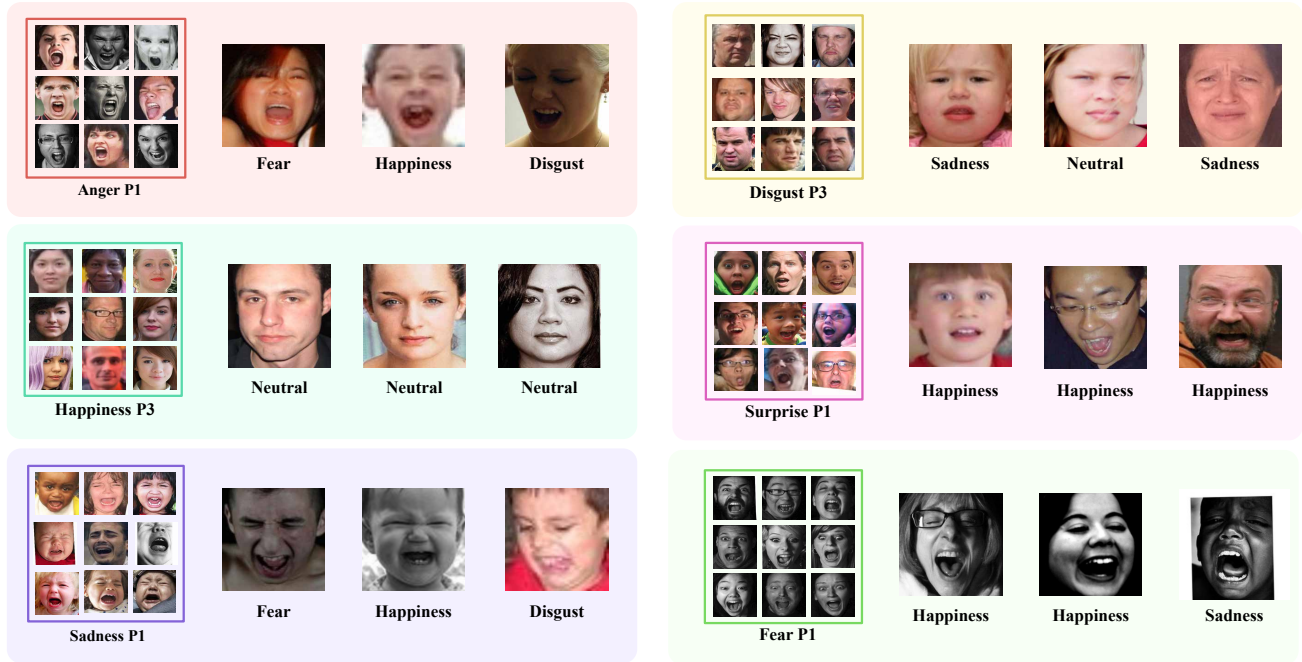


Figure 8. Failure case analysis. For each panel, the left block shows support samples aligned with a specific expression archetype, while the right block shows misclassified samples assigned to this archetype, with their ground-truth labels shown below.

For **Anger P1**, the support samples are characterized by high-intensity open-mouth expressions with strong lower-face tension. The misclassified Fear, Happiness, and Disgust samples assigned to this archetype share a similar forceful mouth-opening pattern, explaining why they are attracted to this anger-related mode despite different labels. **Sadness P1** captures a different structure: many support samples show crying-like expressions with closed or tense eyes and compressed upper-face regions. The misclassified Fear, Happiness, and Disgust samples in this panel exhibit similar crying or strained facial configurations, suggesting that AURA groups them by local visual structure rather than by the coarse class label alone.

For **Fear P1**, the support set mainly contains wide-mouth, high-intensity expressions with limited brow contraction. The misclassified Happiness and Sadness samples assigned to this archetype share similar mouth-stretched configurations, indicating an overlap between fear-like intensity and other open-mouth expressions. In **Disgust P3**, the support samples show nose wrinkling, raised upper lips, and narrowed eyes. The Sadness and Neutral samples assigned to this archetype contain related nose-mouth deformations, suggesting local ambiguity around disgust-like facial cues. In **Happiness P3**, the archetype reflects subtle, low-intensity smiles close to neutral expressions. The Neutral samples assigned to it show mild lip-corner activation and relaxed upper faces, revealing the fuzzy boundary between low-intensity happiness and neutral. Finally, **Surprise P1** captures wide-eyed, open-mouth configurations. The Happiness samples assigned to this archetype share similar surprise-like open-mouth patterns, which explains their attraction to this archetype.

These observations show that AURA’s failure cases are often traceable to specific archetypal modes. Although the final categorical prediction is incorrect, the assigned archetype can still reveal a visually plausible reason for the error. This is useful for model diagnosis: AURA allows users to inspect which semantic mode was activated, what visual pattern it represents, and why a sample was drawn toward that mode. In this sense, failure cases expose the fine-grained structure learned by AURA and highlight the limitations of rigid expression labels for ambiguous or compound facial configurations.

E.2. Broader Failure Modes

Beyond the archetypal explanations shown above, we further analyze failure modes on AffectNet, the largest in-the-wild dataset in our evaluation. Since the overall accuracy on AffectNet-8 is about 65%, we focus on the remaining misclassified samples to understand when AURA fails and what visual factors contribute to these errors.

We observe that a large portion of failures is associated with annotation ambiguity and label noise. Specifically, 28.7% of the test set can be attributed to ambiguous or noisy annotations, consistent with prior studies on in-the-wild facial expression datasets. These cases often arise from subjective judgment, compound expressions, low-intensity affect, or high inter-class visual similarity. AURA is useful in this setting because it not only flags disagreement between the dataset label and the assigned archetype, but also provides a visual explanation through archetype-level facial structure. In many cases, the assigned archetype shares clear local facial geometry with the input, even when its category differs from the ground-truth label.

We also examine pose, illumination, occlusion, and blur. Pose and illumination are not dominant failure sources in our analysis. Even under unusually dark or bright lighting, or under large pose variation with partial facial visibility, AURA can often still match meaningful facial structure. The more severe failures are mainly associated with the loss of recognizable facial geometry. About 4.5% of the test set corresponds to cases where substantial facial structure is missing, such as extreme cropping, foreground occlusion, or cases where only part of the face is visible. Occlusions do not affect the model uniformly: hand occlusions can still be handled when enough facial structure remains visible, while stronger non-facial occlusions, such as subtitles, pacifiers, or masks, can cause failure by removing the geometric cues required for archetype matching.

A further 0.8% of the test set contains extremely blurred samples with almost no discernible facial details. In these cases, even human interpretation is difficult, and the archetype assignment becomes unreliable because the visual evidence needed for matching is largely absent.

Overall, these observations suggest that AURA’s failures are less often caused by ordinary pose or illumination variation, and more often by annotation ambiguity or severe degradation of facial structure. This also clarifies the scope of archetype-based reasoning: AURA can provide meaningful explanations when facial structure is sufficiently visible, but its interpretability depends on the presence of recognizable visual evidence.

F. Ablation Study and Additional Analysis

F.1. Ablation Study

As summarized in Table 8, CLIP with a linear head (Row 1) serves as the baseline. Removing the VAS Projection module (Row 2) forces archetype discovery to operate directly on generic CLIP features, leading to substantial performance drops. Removing the Archetype Contextualization module (Row 3) also degrades performance, showing the importance of modeling local archetype relations. The two discriminative objectives play different roles: removing $\mathcal{L}^{\text{VAS}}$ (Row 4) weakens task-aligned visual projection, while removing archetype-level supervision $\mathcal{L}^{\text{Dis}}$ (Row 5) causes larger drops, indicating that

discriminative archetype semantics are central to stable prediction.

The topology objective is also important. Removing $\mathcal{L}^{\text{Top}}$ (Row 6) degrades all tasks, confirming that geometric organization is necessary for effective archetype learning. The component-level ablations (Rows 7–9) further show that compactness/attraction and separation/repulsion are especially important for categorical and AU tasks, while diversity also contributes strongly to VA regression. For the projection dimension (Rows 10–12), $d = 32$ provides the best overall trade-off. A smaller dimension underfits affective variation, whereas a larger dimension can reintroduce irrelevant visual factors. Finally, the feature-granularity study (Rows 13–14) shows that AffectNet-8 and VA regression favor global features, AU detection benefits from patch-level features, and RAF-DB obtains a small gain from patch-level features.

Table 8. Ablation study of AURA components across four datasets. RAF-DB and AffectNet-8 are reported by accuracy, DISFA by F1 score, and AffectNet-VA by CCC.

Index	Configuration	Removed / Variant	RAF-DB Acc	AffectNet-8 Acc	DISFA F1	AffectNet-VA CCC
0	AURA (full model)	–	94.0	65.2	66.9	74.1
1	CLIP Embedding + Classifier	w/o AURA	85.3	57.3	55.2	63.4
2	Use CLIP Features for Archetype Discovery	w/o VAS Projection	88.1	59.3	58.1	66.2
3	Remove Archetype Contextualization Module	w/o AC Module	92.7	63.1	64.2	73.7
4	No Discriminative Supervision for VAS Embedding	w/o $\mathcal{L}^{\text{VAS}}$	93.5	63.4	63.9	72.2
5	No Discriminative Supervision for Archetypes	w/o $\mathcal{L}^{\text{Dis}}$	89.3	61.6	60.2	69.8
6	No Topology Refinement	w/o $\mathcal{L}^{\text{Top}}$	92.3	62.1	64.3	71.2
7	No Compactness / Score-aware Attraction	w/o $\mathcal{L}_{\text{intra}}/\mathcal{L}_{\text{attr}}$	90.2	61.2	61.7	73.1
8	No Separation / Score-aware Repulsion	w/o $\mathcal{L}_{\text{inter}}/\mathcal{L}_{\text{repu}}$	90.8	62.1	60.0	73.9
9	No Archetype Diversity	w/o $\mathcal{L}_{\text{div}}$	92.7	63.1	65.2	72.3
10	VAS Dimension	16	91.3	60.2	61.5	69.0
11		32	94.0	65.2	66.9	74.1
12		64	93.1	64.9	64.1	73.5
13	Input Feature Granularity	Global-level	94.0	65.2	62.3	74.1
14		Patch-level	94.3	64.7	66.9	73.5

F.2. Sensitivity to the Choice of K_{max}

This section clarifies the distinction between the over-complete archetype upper bound and the effective number of active archetypes used by AURA. We denote the initial upper bound as K_{max} and the stable number of active archetypes as K_{stable} . AURA initializes an over-complete archetype pool to ensure sufficient representational capacity, while the training process identifies a smaller active subset according to the data distribution. Thus, K_{stable} is not directly prescribed by K_{max} , but emerges from the self-organization process.

The adaptivity of AURA is driven by the Self-organized Archetype Discovery module and the Archetype Contextualization module. Together, these objectives encourage informative archetypes to receive consistent assignments while suppressing redundant ones. Let the over-complete archetype pool be $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^{K_{\text{max}}}\}$. During training, AURA maintains a usage counter u_k for each archetype, recording how often $\mathbf{a}^k$ is selected as the nearest archetype. The normalized usage ratio is computed as $\alpha_k = u_k / \sum_{j=1}^{K_{\text{max}}} u_j$. Archetypes with $\alpha_k < \eta_{\text{reset}}$ are regarded as underutilized. During the early stage of optimization, the Archetype Reset Mechanism reinitializes underutilized archetypes to avoid premature collapse of the over-complete pool. After this warm-up stage, the usage statistics stabilize, yielding a consistent active set K_{stable} .

To examine robustness with respect to K_{max} , we conduct a sensitivity analysis on RAF-DB with $K_{\text{max}} \in \{150, 200, 300, 400\}$. As summarized in Table 9, the resulting number of active archetypes remains within a narrow range from 99 to 102 across all settings. Model accuracy varies by at most 0.2%, convergence epochs remain comparable, and the class-wise archetype distribution is highly consistent. Increasing K_{max} mainly increases the cost of maintaining the over-complete archetype pool, while leaving the effective archetype granularity nearly unchanged. These results indicate that AURA is robust to the initial capacity upper bound and that its final representational granularity is governed primarily by data-driven self-organization rather than by a manually fixed prototype count.

F.3. Sensitivity to the Loss-Weight Coefficients λ

This appendix analyzes the sensitivity of AURA to the loss-weight coefficients in the total objective:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{VAS}} \mathcal{L}^{\text{VAS}} + \lambda_{\text{SAD}} \mathcal{L}^{\text{SAD}} + \lambda_{\text{CTX}} \mathcal{L}^{\text{CTX}}.$$

Table 9. Sensitivity of AURA to the choice of $K_{\max}$ on RAF-DB.

$K_{\max}$	K_{stable}	ACC (%)	Convergence Epoch	Archetype Distribution	GFLOPs
150	100	94.1	153	Anger 11 / Disgust 9 / Fear 6 / Happiness 28 / Neutral 19 / Sadness 12 / Surprise 15	0.26
200	99	94.0	147	Anger 11 / Disgust 9 / Fear 6 / Happiness 26 / Neutral 20 / Sadness 12 / Surprise 15	0.29
300	102	94.2	158	Anger 11 / Disgust 10 / Fear 6 / Happiness 29 / Neutral 18 / Sadness 13 / Surprise 15	0.44
400	101	94.1	155	Anger 12 / Disgust 10 / Fear 6 / Happiness 29 / Neutral 19 / Sadness 13 / Surprise 12	0.48

Across all experiments, we use the uniform setting $\lambda_{\text{VAS}} = \lambda_{\text{SAD}} = \lambda_{\text{CTX}} = 1$ without dataset-specific tuning. This simple setting yields strong performance on RAF-DB, EmotioNet, and AffectNet-VA, indicating that AURA does not rely on delicate loss balancing.

This stability comes from the modular structure of AURA. The three objectives primarily target different functional components: $\mathcal{L}^{\text{VAS}}$ aligns projected visual embeddings with task semantics, $\mathcal{L}^{\text{SAD}}$ learns discriminative and topologically organized archetypes, and $\mathcal{L}^{\text{CTX}}$ supervises distance-aware contextualization. Although these objectives are not fully independent, their effects are structurally separated at the module level, which reduces direct gradient competition.

Within $\mathcal{L}^{\text{SAD}}$, the assignment, discriminative, and topology terms play complementary roles. $\mathcal{L}^{\text{Assign}}$ stabilizes feature-archetype alignment, $\mathcal{L}^{\text{Dis}}$ injects task-aware semantics into archetypes, and $\mathcal{L}^{\text{Top}}$ organizes the archetype geometry. For FER, $\mathcal{L}^{\text{Top}}$ enforces compactness, separation, and diversity; for VA/AU, it uses score-aware attraction, score-aware repulsion, and diversity to impose analogous geometric constraints. Since these terms are based on normalized features and comparable task-level scales, the uniform weighting provides a stable default.

To empirically examine sensitivity, we vary one coefficient at a time within $\{0.5, 1.0, 1.5\}$ while keeping the other two fixed at 1.0. As shown in Table 10, final performance remains stable across all settings, with variations of at most 0.5 on RAF-DB, 0.5 on EmotioNet, and 0.3 on AffectNet-VA. In contrast, convergence epochs vary more noticeably, suggesting that loss weights mainly affect optimization speed rather than the final model quality. These results support the robustness of AURA to moderate changes in loss-weight coefficients.

Table 10. One-factor sensitivity analysis of the loss-weight coefficients across FER (RAF-DB), AU detection (EmotioNet), and VA regression (AffectNet-VA). One coefficient is varied at a time while the other two are fixed at 1.0. The gray row denotes the default setting.

$(\lambda_{\text{VAS}}, \lambda_{\text{SAD}}, \lambda_{\text{CTX}})$	RAF-DB Acc (%)	Epoch	AU-F1 (%)	Epoch	VA-CCC (%)	Epoch
(1.0, 1.0, 1.0)	94.0	153	67.3	136	74.1	188
(0.5, 1.0, 1.0)	93.8	187	67.1	165	74.1	203
(1.5, 1.0, 1.0)	94.0	149	66.8	138	74.2	184
(1.0, 0.5, 1.0)	94.2	176	67.1	148	73.9	189
(1.0, 1.5, 1.0)	94.3	142	67.2	136	74.0	196
(1.0, 1.0, 0.5)	93.8	180	66.9	164	74.0	207
(1.0, 1.0, 1.5)	94.1	146	67.2	122	74.2	175

F.4. Sensitivity to the Topology Margin

Table 11. Sensitivity analysis of the topology margin m_{top} across FER (RAF-DB), AU detection (EmotioNet), and VA regression (AffectNet-VA). For FER, m_{top} corresponds to the inter-class similarity margin γ in $\mathcal{L}_{\text{inter}}$; for VA/AU, it corresponds to the distance margin δ in the score-aware topology losses, with the score threshold ϵ kept fixed.

m_{top}	RAF-DB Acc (%)	EmotioNet AU-F1 (%)	AffectNet-VA CCC (%)
0.1	93.2	66.7	73.9
0.2	93.9	67.2	74.1
0.3	94.1	67.3	74.0
0.4	94.0	67.3	73.8
0.5	93.6	66.9	73.4

This appendix analyzes the sensitivity of AURA to the topology margin m_{top} . Since AURA uses task-specific topology objectives, m_{top} is used here as a generic sensitivity-control variable for the corresponding margin in each setting. For FER,

m_{top} corresponds to the inter-class similarity margin γ in $\mathcal{L}_{\text{inter}}$. For VA regression and AU detection, m_{top} corresponds to the distance margin δ in the score-aware attraction and repulsion losses. The score threshold ϵ , which determines whether two archetypes form an attraction pair or a repulsion pair according to their predicted score difference, is kept fixed throughout this sensitivity analysis.

In the discrete FER setting, the margin controls how strongly different expression regions are separated. The inter-class separation loss penalizes class-center pairs whose cosine similarity exceeds γ :

$$\mathcal{L}_{\text{inter}} = \frac{1}{|\mathcal{S}_{\text{inter}}|} \sum_{(c, c') \in \mathcal{S}_{\text{inter}}} \max(0, \cos(\tilde{\mu}_c, \tilde{\mu}_{c'}) - \gamma),$$

where $\mathcal{S}_{\text{inter}}$ denotes the set of distinct class-center pairs and $\tilde{\mu}_c$ is the normalized center of class c . Since cosine similarity between normalized centers lies in $[-1, 1]$, a positive margin in a moderate range imposes separation between different expression regions without forcing the class centers to become overly dispersed.

In the continuous VA/AU setting, the role of the margin is different. The score threshold ϵ first determines whether two archetypes have similar or distinct predicted affective scores. The distance margin δ then controls the geometric tolerance in the hinge losses. Specifically, the attraction term penalizes similar-score archetypes when their cosine distance exceeds δ , while the repulsion term penalizes distinct-score archetypes when their cosine distance falls below δ . Thus, δ does not define the score boundary between similar and dissimilar pairs. It only controls the local geometric radius used to organize the continuous archetype manifold.

As shown in Table 11, AURA remains stable across a moderate range of topology margins. The best or near-best results are obtained around $m_{\text{top}} \in [0.2, 0.4]$: RAF-DB reaches 94.1 at $m_{\text{top}} = 0.3$, EmotionNet reaches 67.3 at both $m_{\text{top}} = 0.3$ and 0.4, and AffectNet-VA reaches 74.1 at $m_{\text{top}} = 0.2$. Performance drops slightly at the extreme settings, especially when $m_{\text{top}} = 0.1$ or 0.5, indicating that overly weak or overly restrictive topology constraints can reduce the quality of the learned archetype geometry. Overall, these results show that AURA is robust to moderate changes in the topology margin and does not require careful tuning of this hyperparameter.

F.5. Global-Level vs. Patch-Level Features

We ablate the input feature granularity in AURA by comparing *global-level* and *patch-level* representations, as reported in Table 8. The results show a task- and dataset-dependent pattern. Patch-level features yield a slight improvement on RAF-DB (94.3 vs. 94.0) and a substantial gain on DISFA (66.9 vs. 62.3), while global-level features perform better on AffectNet-8 (65.2 vs. 64.7) and AffectNet-VA (74.1 vs. 73.5). This behavior is consistent with the different visual evidence required by each task and dataset.

Why patch-level features help on RAF-DB and AU detection. RAF-DB images are relatively well aligned and centrally framed, so local facial evidence is often clean and consistently observable. Many expression categories contain salient local cues, such as brow contraction, nose-region deformation, eyelid widening, or mouth-shape changes. Patch-level features can emphasize these localized patterns, leading to a small improvement on RAF-DB. This effect is much stronger for AU detection, where the task directly depends on localized facial muscle activations. The large gain on DISFA indicates that patch-level features provide more precise evidence for detecting local AU patterns than a single global representation.

Why global features are better on AffectNet-8 and AffectNet-VA. AffectNet contains larger variations in pose, illumination, expression intensity, image quality, and annotation ambiguity. Under these conditions, isolated local patches can be less reliable, since some facial regions may be misaligned, occluded, or visually corrupted. Global features aggregate information across the whole face and are therefore more robust for in-the-wild expression recognition on AffectNet-8. This effect is also clear in AffectNet-VA, where valence and arousal describe holistic affective states rather than isolated local activations. Global representations better preserve the overall facial configuration needed for continuous VA regression.

The ablation shows that AURA is flexible with respect to input feature granularity. Patch-level representations are beneficial when reliable local evidence is central to the task, as in AU detection and the relatively clean RAF-DB setting. Global representations are preferable when holistic affective context and robustness to in-the-wild variation are more important, as in AffectNet-8 and AffectNet-VA. Given the small gap on RAF-DB and the lower computational cost of global features, we use the global-level representation as the default for FER and VA, while retaining patch-level features for AU detection.

F.6. Encoder-Agnostic Behavior of AURA

To examine whether AURA’s gains arise from the proposed archetype mechanism rather than solely from CLIP’s vision-language priors, we extend the evaluation to both **CLIP** (Radford et al., 2021) and **DINO** (Caron et al., 2021). DINO is a self-supervised visual backbone without text-image alignment, making it a useful test case for assessing whether AURA can operate beyond CLIP’s semantically aligned feature space. This analysis helps isolate the contribution of archetype-based structuring from encoder-specific pretraining effects.

We evaluate each encoder under two configurations. For DINO, we first consider **DINO FT**, where the full DINO visual encoder is fine-tuned jointly with task-specific heads. We then construct **AURA-DINO**, where the CLIP backbone in AURA is replaced by a pretrained DINO encoder, while the encoder remains *frozen*. In this setting, DINO is used only as a feature extractor, and only AURA’s trainable modules are optimized.

For completeness, we apply the same comparison to CLIP. In the **CLIP FT** setting, the CLIP visual encoder is fine-tuned end-to-end with task-specific heads. In **AURA-CLIP**, the pretrained CLIP encoder is kept *frozen*, and AURA operates on top of its fixed visual embeddings. This cross-encoder comparison allows us to test whether AURA’s improvements are tied to CLIP’s vision-language alignment or remain effective under a purely visual self-supervised encoder.

As shown in Table 12, AURA consistently improves over the corresponding fully fine-tuned baselines for both CLIP and DINO. Notably, AURA-DINO, despite using a frozen encoder, outperforms DINO FT by +4.3% accuracy on RAF-DB, +6.6 CCC points on AffectNet-VA, and +4.7 F1 points on EmotionNet. This indicates that AURA’s gains do not rely solely on CLIP’s multimodal alignment, since similar improvements are observed with a self-supervised visual backbone.

AURA-CLIP achieves the strongest absolute performance, which is expected given CLIP’s stronger semantic priors. Its gains over CLIP FT are also slightly larger than those observed for DINO, suggesting that AURA can benefit from stronger pretrained representations while still providing improvements beyond encoder adaptation. These results support the **encoder-agnostic** behavior of AURA across the evaluated backbones: the archetype mechanism can structure both vision-language and self-supervised visual representations by discovering semantic anchors and decomposing fine-grained intra-class variation.

Table 12. Evaluation of AURA with CLIP and DINO encoders. “FT” denotes full fine-tuning, and “AURA” denotes a frozen encoder with AURA modules.

Method	Encoder	RAF-DB Acc (%)	AffectNet-VA CCC (%)	EmotionNet AU-F1 (%)
CLIP FT	CLIP (fine-tuned)	89.1	66.4	62.3
AURA-CLIP	CLIP (frozen)	94.0 (+4.9)	74.1 (+7.7)	67.3 (+5.0)
DINO FT	DINO (fine-tuned)	88.3	65.4	60.7
AURA-DINO	DINO (frozen)	92.6 (+4.3)	72.0 (+6.6)	65.4 (+4.7)

F.7. Cross-Dataset Generalization

To examine whether the learned archetypes capture transferable affective structure rather than only dataset-specific statistics, we conduct cross-dataset evaluation between AffectNet-7 and RAF-DB. We evaluate two transfer directions: training on AffectNet-7 and testing on RAF-DB, and training on RAF-DB and testing on AffectNet-7. This setting is challenging because the two datasets differ in image distribution, annotation style, and expression composition.

We compare AURA with strong FER baselines and representative test-time adaptation (TTA) methods. The TTA results are reported under the same cross-dataset setting following prior evaluation protocols (Turnbull et al., 2026). Importantly, AURA is evaluated without target-domain test-time adaptation, while TTA methods are allowed to adapt to the target-domain test distribution.

As shown in Table 13, AURA achieves the best performance in both transfer directions. On AffectNet-7 → RAF-DB, AURA reaches 79.4%, outperforming SE-ResNet, MHAN, POSTER++, and all TTA baselines. On RAF-DB → AffectNet-7, AURA obtains 51.7%, again exceeding the strongest compared methods. This is notable because several TTA baselines are allowed to adapt to the target-domain test distribution, whereas AURA relies only on the learned archetype structure.

These results suggest that AURA does not merely memorize dataset-specific label statistics. The learned archetypes provide a structured visual representation that remains useful under cross-dataset distribution shifts. While this does not imply

Table 13. Cross-dataset evaluation between AffectNet-7 and RAF-DB. AURA is evaluated without test-time adaptation.

Method	AffectNet-7 $\rightarrow$ RAF-DB	RAF-DB $\rightarrow$ AffectNet-7
SE-ResNet (Huang et al., 2023)	77.4	42.6
MHAN (Wang et al., 2025)	75.6	46.3
POSTER++ (Mao et al., 2025)	74.0	48.5
TENT (Wang et al., 2020)	74.8	49.0
EATA (Niu et al., 2022)	72.3	49.4
SAR (Niu et al., 2023)	74.8	48.9
T3A (Iwasawa & Matsuo, 2021)	74.3	49.1
NOTE (Gong et al., 2022)	72.3	49.6
RoTTA (Yuan et al., 2023)	72.1	49.5
AURA (Ours)	79.4	51.7

universal transfer across all affective datasets, it provides evidence that archetype-based organization captures reusable facial affective structure beyond a single training dataset.

F.8. End-to-End Efficiency Analysis

AURA is designed to operate in two practical modes. In the cached-feature mode, frozen visual features are extracted once and AURA is trained or evaluated at the feature level. This is the setting used for our main efficiency comparison, since the CLIP encoder is frozen and does not need to be repeatedly optimized. To clarify the full computational profile, we additionally report end-to-end measurements that include frozen feature extraction.

All measurements are conducted on a workstation with a single NVIDIA A100 40GB PCIe GPU, an Intel Xeon Gold 6330 CPU, 256GB RAM, CUDA 12.1, and PyTorch 2.1. Table 14 reports total parameters, FLOPs, latency per image, and peak GPU memory under different configurations. The results show that the trainable AURA modules are lightweight: feature-level training requires only 3.30M trainable parameters and 0.26G FLOPs, while inference on cached features takes only 0.007 ms per image. When online inference is required, the total cost includes both frozen CLIP feature extraction and AURA inference, resulting in 152.55M parameters and 22.06G FLOPs. This clarifies that the low-overhead claim refers to the feature-level AURA modules after frozen feature extraction, while the end-to-end online cost is dominated by the frozen visual encoder.

Table 14. Configuration-wise efficiency analysis of AURA. Feature-level training and cached-feature inference report the cost of trainable AURA modules after frozen feature extraction. Online inference includes both frozen CLIP feature extraction and AURA inference.

Configuration	Total Params (M)	FLOPs (G)	Latency / image (ms)	Peak GPU Memory (GB)
Frozen feature extraction	149.25	21.80	0.410	4.32
Feature-level training	3.30	0.26	0.018	1.21
AURA inference on cached features	3.30	0.26	0.007	0.56
Online inference	152.55	22.06	0.440	4.67

These results provide a clearer view of AURA’s efficiency. If features are cached, AURA adds only a small computation and memory footprint. If the full frozen encoder is included, the online inference cost remains comparable to a standard frozen-encoder pipeline, with AURA adding only a lightweight projection, archetype matching, and task head. Therefore, AURA’s efficiency advantage comes from avoiding backbone fine-tuning and reducing inference after feature extraction to simple feature-level operations.

G. Scope, Limitations, and Responsible Use

G.1. Scope and Limits of the Archetype Paradigm

The scope of AURA lies in a shared archetype-learning principle. Across FER, AU detection, and VA regression, the framework projects visual features into an archetype space, assigns samples to learned archetypes, shapes their topology with task-aware objectives, and refines local archetype relations during training. Each task instantiates this principle according to its output geometry: global features and class-based compactness and separation support discrete expression recognition, patch-level features support localized multi-label AU detection, and score-aware relations support continuous VA regression. Under this formulation, AURA’s robustness and interpretability are assessed at the framework level, where archetypes

provide structured visual evidence for model diagnosis, dataset auditing, and failure analysis. These explanations describe the model’s visual evidence, not universal or causal mechanisms of human emotion.

G.2. Responsible Use and Privacy Considerations

AURA operates on facial images and derived visual representations, so its use requires careful privacy protection. Raw images, frozen visual embeddings, learned archetypes, and assignment traces may all contain sensitive information when derived from human faces. They should be handled with appropriate access control, encryption, retention limits, and deletion policies. Dataset licenses, consent conditions, and access restrictions should also extend to trained models, cached embeddings, and archetype codebooks derived from those data.

Real-world deployment further requires explicit consent, domain-specific validation, and human oversight. AURA is evaluated for affective representation learning, not for non-consensual profiling, behavioral monitoring, surveillance, or high-stakes decision-making. Its predictions should not be used as standalone evidence for mental health diagnosis, employment decisions, legal assessment, education monitoring, or other consequential judgments.

AURA’s interpretability should be understood within its proper scope. The learned archetypes and decision trails reveal the visual evidence used by the model, such as resemblance to facial patterns involving jaw drop, brow contraction, or eye widening. They do not establish a person’s internal affective state or the cause of an emotion. Facial appearance is context-dependent, and similar configurations may carry different meanings across individuals, cultures, demographic groups, and recording conditions. Responsible use therefore requires subgroup-level evaluation, calibration analysis, local validation, and clear communication of the limits of facial affect inference.